MadLoop with TIRust

One-loop reduction · registered campaign report

CutTools · Ninja · COLLIER · NinjaRS · TIRust

Starting MadLoop revision `9228fc58e8f8` · full helicity/color sums

Separate sources. Main report and plots: `2c13246ffa63` (66 lanes). Source `0fcd2cd873d6`: 11 cases, 66 selected lanes in the supplement. Historical results and guard-only timings do not qualify the changed source.

Selected campaign: `invariant-symmetry-1`. Selected numerical identity: `f217762db032b8b7`.

Registered exporter anchor: `2c13246ffa63`. Observed formal paired-timing export revisions: not yet measured. Documentary revisions can differ while the registered exporter bytes remain identical.

Complete delivery qualification: incomplete. 11 of 11 physical cases have closed: 66 lane slots × 100 saved points. 11 complete cases; 11 pass 3e-6. Current numerical status: **passed**. The 1e-8 target is missed by 3 complete cases; all lane outcomes remain visible. The eleven-process numerical matrix, controls, existing-backend regressions, precision cohorts, implementation checks and unresolved stress cases retain separate statuses.

Precise performance acceptance: deferred to Hua-Sheng. Final timings will run on his local machine. Shared-host timings are provisional, without CPU reservation, and cannot establish the ≤1.05 upper confidence bound. This snapshot contains no new paired timing samples.

Current-source numerical checkpoint. This report selects `invariant-symmetry-1`. Earlier shared-host timings remain in the archived 19-page split-dp-terminal-1 report, bound to commit `584dae7875d2`. Those timings retain their historical source identity. Current performance gates remain deferred.

Runtime colour: green = faster, white = parity, red = slower. All displayed ratios are candidate / reference; smaller is faster.

Table 1: Main process matrix. Timings include the actual rescue work. Physics and runtime acceptance are separate.

Process · correction	Physics	Baseline µs/point	TIRust chain µs/point	Ratio · 95% interval	Existing-backend regressions
$u\bar{u} \to Z$ QCD · 2→1 · 100 points	pass	—	—	— deferred	deferred
$gg \to t\bar{t}$ QCD · 2→2 · 100 points	pass	—	—	— deferred	deferred
$u\bar{u} \to Zgg$ QCD · 2→3 · 100 points	pass	—	—	— deferred	deferred
$u\bar{u} \to ZZgg$ QCD · 2→4 · 100 points	pass	—	—	— deferred	deferred
$u\bar{d} \to W^+$ EW · 2→1 · 100 points	pass	—	—	— deferred	deferred
$e^+e^- \to \mu^+\mu^-$ EW · 2→2 · 100 points	pass	—	—	— deferred	deferred
$e^+e^- \to t\bar{t}\gamma$ EW · 2→3 · 100 points	pass	—	—	— deferred	deferred
$e^+e^- \to \mu^+\mu^-\gamma\gamma$ EW · 2→4 · 100 points	pass	—	—	— deferred	deferred

This study measures the new integration against the untouched starting branch, which already includes NinjaRS and earlier Fortran optimizations. It does not attribute those earlier improvements to TIRust.

1 Retained predecessor failures

The following qualified failures belong to earlier candidate implementations. They remain authoritative for those numerical sources and fixed policies. Results for the selected campaign appear in the main matrix; a new campaign does not replace or erase a previous candidate's failure.

Table 2: All retained qualified ancestor failures, including discovered and unselected attempts. Source identities and every read evidence file are bound in the report build manifest.

Process · comparison	Previous candidate source	Retained ratio · 95% interval	Retained failure evidence
$u\bar{u} \to Z$ `tirust-first`	`legacy` revision `c91cb85f3aa9` source SHA `74116e4bb2ba7abd`	1.12105 1.10349–1.13889 **fail**	`runs/qcd_n1_uux_z/tirust-first-attempt-3`

2 Five-backend numerical agreement

The same saved momenta, model parameters, coupling orders and scale are evaluated by all five tools. The regression criterion is `3e-6`, with `1e-8` as the ordinary-point target. Born, finite, simple-pole and double-pole coefficients are retained. Near-zero poles use the existing MadLoop comparison policy. Final accepted DP/QP outcomes and failed evaluations are retained; the histograms do not enumerate internal rescue attempts.

This table reparses the final complete steady warmup traversal: 100 saved points, baseline CutTools and five candidate reducers. It is untimed numerical evidence. It supplies neither paired runtime acceptance nor native-DP cohort timings. Internal retry counters and hardware FPE flags are **unobserved**. This warmup-only numerical traversal finishes before unit 6 is redirected, so its console diagnostics are retained. The numerical parser does not attribute or gate every console warning per call. Subsequent post-READY timed RUN batches suppress unit 6 output.

Process · saved points	CutTools	Ninja	COLLIER	NinjaRS	TIRust
$u\bar{u} \to Z$ main · 100 expected Baseline CT: **complete**	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass
$gg \to t\bar{t}$ main · 100 expected Baseline CT: **complete**	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass
$u\bar{u} \to Zgg$ main · 100 expected Baseline CT: **complete**	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: miss	**pass** DP 100 / QP 0 1e-8: miss	**pass** DP 100 / QP 0 1e-8: miss	**pass** DP 87 / QP 13 1e-8: miss
$u\bar{u} \to ZZgg$ main · 100 expected Baseline CT: **complete**	**pass** DP 88 / QP 12 1e-8: pass	**pass** DP 97 / QP 3 1e-8: miss	**pass** DP 100 / QP 0 1e-8: miss	**pass** DP 96 / QP 4 1e-8: miss	**pass** DP 14 / QP 86 1e-8: miss
$u\bar{d} \to W^+$ main · 100 expected Baseline CT: **complete**	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass
$e^+e^- \to \mu^+\mu^-$ main · 100 expected Baseline CT: **complete**	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass
$e^+e^- \to t\bar{t}\gamma$ main · 100 expected Baseline CT: **complete**	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass
$e^+e^- \to \mu^+\mu^-\gamma\gamma$ main · 100 expected Baseline CT: **complete**	**pass** DP 93 / QP 7 1e-8: pass	**pass** DP 100 / QP 0 1e-8: miss	**pass** DP 100 / QP 0 1e-8: miss	**pass** DP 100 / QP 0 1e-8: miss	**pass** DP 40 / QP 60 1e-8: miss
$u\bar{u} \to g\gamma$ control · 100 expected Baseline CT: **complete**	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass
$gg \to H$ control · 100 expected Baseline CT: **complete**	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass
$gg \to HH$ control · 100 expected Baseline CT: **complete**	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass

Every available comparison uses baseline CutTools rows on exactly matching point and split-order indices. A failed baseline leaves complete candidate lanes uncompared; their DP/QP counts remain observed. The separate 1e-8 target is not substituted for the 3e-6 regression criterion. Failed, incomplete and corrupt observations retain their diagnostics and never become successful rows.

Agreement is assessed using independent numerical backends and higher precision when a discrepancy occurs. Known difficult scalar-box regions are retained as stress cases. A backend's own success flag does not establish cross-backend agreement.

3 Completed physical pilot checks

These eight separately retained checks each evaluate 100 saved physical points with all five reducers. They use the common strict stability policy and pass the `3e-6` regression criterion. Their timings are ineligible for performance acceptance, and they do not fill the pending timing-campaign rows above. An independent audit reparsed 4,800 evaluations and checked the retained inputs, executable provenance and raw statuses.

> These historical pilots retain their original candidate sources in the pilot index. They do not establish numerical acceptance for the selected campaign `invariant-symmetry-1`.

Table 3: Historical physical pilots, independent of the runtime gate. Residuals exclude poles accepted by the documented small-pole exception. All four existing reducers finally accepted 100 DP results in each row.

Process · 100 points	Five-backend regression	Ordinary target	Largest symmetric residual	TIRust accepted DP / QP
$u\bar{u} \to Z$	**pass**	**pass**	2.69×10^{-13}	100 / 0
$gg \to t\bar{t}$	**pass**	**pass**	4.58×10^{-12}	100 / 0
$u\bar{u} \to Zgg$	**pass**	**miss**	2.81×10^{-6}	87 / 13
$u\bar{d} \to W^+$	**pass**	**pass**	1.95×10^{-12}	100 / 0
$e^+e^- \to \mu^+\mu^-$	**pass**	**pass**	2.65×10^{-12}	100 / 0
$u\bar{u} \to g\gamma$	**pass**	**pass**	2.61×10^{-11}	100 / 0
$gg \to H$	**pass**	**pass**	1.55×10^{-10}	100 / 0
$gg \to HH$	**pass**	**pass**	4.61×10^{-9}	100 / 0

Historical split-dp-terminal-1 QCD 2→3 adjudication. The original comparison against CutTools DP misses the `1e-8` target at 28 points. CutTools, Ninja and TIRust QP agree within `9.89e-11` on four diagnostic points. Across all 28 target misses, the saved TIRust and COLLIER results meet the target against CutTools QP; CutTools DP misses all 28, while Ninja and NinjaRS miss only point 39. Every comparison still passes the regression criterion. These fixed-QP diagnostics are explicitly unchecked and do not establish automatic rescue stability.

The index `experiments/numerical/completed-pilots-index.json` binds 491 retained files and preserves the original comparisons. The detailed QP adjudication is in `experiments/qcd-n3-adjudication/`. Final accepted precision does not enumerate every internal reduction attempt. The remaining three large-process pilots are outside this frozen eight-case index.

4 Absolute timing by reduction backend

Values below are warmed full-amplitude times in µs per point. Initialization, compilation and model generation are excluded and recorded separately. All tools use matching pole, helicity/color and stability settings. Cache behaviour and actual rescue work remain part of the measurement. Recorded backend/precision histograms identify the final accepted reducer for every timed amplitude; they do not count every internal rescue attempt. Control-process speeds are descriptive; the ≤1.05 runtime gate applies to the eight main processes.

Table 4: Per-backend timing observations. Missing measurements are not represented as zero.

Process	CutTools	Ninja	COLLIER	NinjaRS	TIRust
$u\bar{u} \to Z$	—	—	—	—	—
$gg \to t\bar{t}$	—	—	—	—	—
$u\bar{u} \to Zgg$	—	—	—	—	—
$u\bar{u} \to ZZgg$	—	—	—	—	—
$u\bar{d} \to W^+$	—	—	—	—	—
$e^+e^- \to \mu^+\mu^-$	—	—	—	—	—
$e^+e^- \to t\bar{t}\gamma$	—	—	—	—	—
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	—	—	—	—	—
$u\bar{u} \to g\gamma$	—	—	—	—	—
$gg \to H$	—	—	—	—	—
$gg \to HH$	—	—	—	—	—

5 Initialization and accepted precision

Each cell gives the first-amplitude initialization plus complete-stream warmup time in seconds, followed by the fraction of timed amplitudes finally accepted in QP. Process startup and parameter-card loading are outside the cold interval. A DP-only label refers to accepted amplitudes; it does not assert that no unsuccessful internal attempt occurred. The warmed timings on the previous page include all rescue work.

Table 5: Cold intervals and final accepted precision. Exact backend/precision counts and point-level results accompany the raw observations.

Process	CutTools	Ninja	COLLIER	NinjaRS	TIRust
$u\bar{u} \to Z$	—	—	—	—	—
$gg \to t\bar{t}$	—	—	—	—	—
$u\bar{u} \to Zgg$	—	—	—	—	—
$u\bar{u} \to ZZgg$	—	—	—	—	—
$u\bar{d} \to W^+$	—	—	—	—	—
$e^+e^- \to \mu^+\mu^-$	—	—	—	—	—
$e^+e^- \to t\bar{t}\gamma$	—	—	—	—	—
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	—	—	—	—	—
$u\bar{u} \to g\gamma$	—	—	—	—	—
$gg \to H$	—	—	—	—	—
$gg \to HH$	—	—	—	—	—

6 Full versus poles-only tensors

10 of 12 predeclared settings completed seven alternating full/poles-only timing pairs; two DP settings were refused in both modes. All returned requested pole coefficients agree. These warmed tensor diagnostics use the historical release, with numerical caches disabled. They do not qualify full-amplitude performance.

Table 6: Median wall µs per call and geometric mean paired ratios. Green favors poles-only; red favors full evaluation. Intervals use seven paired log ratios and Student-t with six degrees of freedom.

Tensor · N / rank	Route	Prec.	Full µs	Poles µs	Poles / full · 95% interval
Massive tadpole N=1 · rank 6	Public Fortran	DP	9.01	9.42	**×1.041** 1.032–1.049
Massive tadpole N=1 · rank 6	Public Fortran	QP	1828	1838	**×1.005** 0.999–1.011
Massive bubble N=2 · rank 4	Generated adapter	DP	11.8	11.3	**×0.951** 0.939–0.964
Massive bubble N=2 · rank 4	Generated adapter	QP	1738	1121	**×0.646** 0.644–0.649
Infrared triangle N=3 · rank 3	Generated adapter	DP	8.95	8.78	**×0.977** 0.954–1.000
Infrared triangle N=3 · rank 3	Generated adapter	QP	1038	858	**×0.823** 0.815–0.832
Massive box N=4 · rank 4	Generated adapter	DP	32.2	18.0	**×0.558** 0.551–0.564
Massive box N=4 · rank 4	Generated adapter	QP	15384	2220	**×0.144** 0.141–0.146
Massive pentagon N=5 · rank 2	Generated adapter	DP	—	—	Both modes refused
Massive pentagon N=5 · rank 2	Generated adapter	QP	39078	2177	**×0.056** 0.055–0.057
Massive hexagon N=6 · rank 2	Generated adapter	DP	—	—	Both modes refused
Massive hexagon N=6 · rank 2	Generated adapter	QP	84852	5144	**×0.061** 0.060–0.061

Scope and observed limits. The rank-six tadpole calls the public Fortran bridge; the other tensors use the existing generated QCD 2⟶4 adapter, whose rank capacity is five. No generated rank limit was changed. The DP tadpole is about 4% slower in poles-only mode; QP is near parity. The massive N=5 and N=6 fixtures have zero poles, so their large QP gains do not describe infrared-singular pentagons or hexagons. DP refusals carry no timing claim.

Numerical caches are off; structural plans and scratch remain warm. Each measured batch lasted 0.825–1.925 s on pinned CPU 9. Driver tracing and coefficient output are outside the timer; input restoration, adapter mapping and bridge serialization remain inside. Shared-host compilation was active; host-load snapshots and CPU times are retained. These timings are explicitly ineligible for the full-amplitude acceptance gate.

Requested poles match within the declared DP/QP bounds; the largest scaled residual is `3.47e-18`. Omitted finite slots are NaN in the generated adapter and zero placeholders in the public bridge. The historical DSO is pinned by SHA256 `49af12f0b882…cd71ca12` and subsequent builds are outside this measurement. Raw batches, exact identities, and the independently audited 79-file inventory are retained under `experiments/poles-timing/`.

7 Methods and qualification

7.1 Physical samples and timing protocol

The main matrix contains QCD and electroweak corrections at every multiplicity from 2⟶1 through 2⟶4, using `loop_sm` and `loop_qcd_qed_sm_Gmu`. Each main case declares 100 deterministic points. On-shell 2⟶1 points vary their boost at the physical invariant; higher multiplicities use 1 TeV. Mixed QCD/QED and loop-induced cases supplement the main matrix. Threshold, soft/collinear, small-Gram and complex-mass cases are separate stress controls.

Performance acceptance uses at least seven alternating paired repetitions. The minimum batch target is one second; calibration aims for two seconds to accommodate shared-host speed variation. Actual durations are retained, and every batch must reach 95% of the minimum. The starting revision and candidate share their compiler, library installations, model/cards, point order and physical settings. Timing begins after the campaign's builds finish and uses one allowed CPU. The upper 95% paired ratio bound must be ≤1.05 separately for every main process. Averages cannot hide a failing row. A FULL–SKIP quantity, when reported in supporting data, is an **estimated reduction stage** rather than a pure tensor-kernel stopwatch.

All timing attempts are retained. Only an objectively short series with passing numerical checks can be retried within the same candidate campaign. A qualified runtime failure remains authoritative; a later favorable repetition cannot erase it. Source optimizations require a new candidate identity and campaign.

The loop-induced controls retain their generated default chain `7|6|1` their prescribed TIRust-first chain is `9|7|6|1`. These are configured chains, not measurements of which reducer ultimately supplied a result.

The recorded cold interval includes first-amplitude initialization and warming the complete saved point stream; it excludes process startup and parameter-card loading. Numerical objects and linked archives are bound to the retained compiler recipes and hashes. The benchmark runs on a shared host; host-load observations accompany timing pairs, and this study does not claim exclusive use of the machine.

The common benchmark card sets `MLStabThres=1e-6` for both revisions and all reducers, with independent stability checks enabled. This policy was frozen before acceptance measurements after a native stress audit exposed correlated errors below the original threshold. Production retains its `1e-3` default. The tighter estimator does not replace the independent numerical truth gate or guarantee detection of every error.

7.2 Interface, poles and caches

MadLoop selects TIRust explicitly as `MLReductionLib=9`. A Fortran adapter transfers supplied invariants, complex squared masses and the squared renormalization scale to a separate Rust bridge. Tensor moments are permuted into MadLoop order and contracted in Fortran. The quadruple-precision path preserves binary128 inputs and uses MPFR working precision; double-double arithmetic is not labelled quadruple precision.

`LoopPoleMode` selects bare-loop UV/IR residues while preserving the ordinary finite coefficient. The distinct poles-only diagnostic path omits finite scalar roots and finite tensor arithmetic; it preserves double-pole contributions to simple poles and scaleless UV/IR separation. Partial results cannot be used as complete physical amplitudes. Cache and precision stores are separated by evaluation mode, and stability rechecks use independent values.

> Detailed interface, cache, QP, poles-only, and relocation qualification records have not yet been incorporated in this data release. Their absence is not a passing result.

7.3 Earlier measured Fortran change

The selected optimization computes ordered momentum prefixes once, retaining the original zero seed and addition order. For six propagators it reduces prefix additions from 60 to 20. At each of `-O0` and `-O3`, 481,536 comparisons passed bitwise output and IEEE-exception checks across DP, QP and four rounding modes. Isolated six-propagator kernel median paired ratios were 0.443/0.347 (DP/QP, `-O0`) and 0.703/0.356 (`-O3`). These are shared-host microbenchmarks, not physical-amplitude speedups. Full-wrapper measurements were smaller and less consistent. Raw observations and source are retained in `experiments/fortran-prefix/`.

7.4 Retained numerical limitations

The bridge's 66 DP/QP stress evaluations include 14 explicit DP refusals and one served DP tensor coefficient above the diagnostic `3e-6` criterion. All 33 QP evaluations pass. For the failing massive rank-five box, native MadLoop reversal checks at the default `MLStabThres=1e-3` can accept correlated DP errors. At `1e-6`, the actual production rescue block with real tensor callbacks switches to QP and resolves the discrepancy below `1.2e-15`. The fixture, failing components,

native check values and QP adjudication remain under `qualification/tirust-bridge/stability-audit/`. This isolated control-flow test omits physical helicity/filter initialization and does not replace matrix-element comparisons.

Pole-only tests prove that finite evaluators and finite tensor arithmetic are skipped. This does not imply every scalar topology is faster: illustrative warm scalar measurements include slower simple massless triangles and boxes, alongside faster massive and soft examples. A completely collapsed massless scalar box returns an explicit error. No unsupported request is counted as accelerated execution.

7.5 Reproduction and provenance

The accompanying `campaigns/invariant-symmetry-1/cases.json` fixes the matrix and thresholds; `campaigns/invariant-symmetry-1/summary.json` is derived from retained observations. The selected manifest is `campaigns/invariant-symmetry-1/manifest.json`. Its ancestor summary snapshots and qualified failures retain the previous candidates' identities separately. The source plan, baseline manifest, dependency revisions, build environment, and raw measurement paths accompany this report. Rebuilding the PDF does not rerun numerical measurements.

Component	Recorded environment
Baseline	`9228fc58e8f8e48dadc70a6fcfb6617845aaeb1f`
Rust	rustc 1.97.1 (8bab26f4f 2026-07-14) (built from a source tarball)
Fortran	GNU Fortran (GCC) 15.2.0
Typst	typst 0.15.1 (9dfd3a08)
TIRust	`854b79dcb7fffcb35e92b4a8a3dd443369b798c4`
oneloopmaster	`bd703d09a78f7849a6b86a07096e6f222fc01193`

Presentation follows the pyAmpliCol report at revision 7266c1600f6e: its page geometry, fonts, navy headings, compact tables and logarithmic timing palette. All numerical measurements and conclusions in this report belong to the present MadLoop integration.

8 Timing eligibility and control-process scope

TODO — Hua-Sheng, local machine. Rebuild the delivered candidate and baseline with identical pinned inputs; replay all 100 saved points; then run at least seven alternating single-core pairs with approximately one-second batches. Report initialization and actual backend/precision use separately. The upper 95% ratio bound ≤1.05 must be checked for every main chain and existing-backend comparison. Shared-host work is best effort without reserving CPUs; it cannot pass this final performance requirement.

User amendment: `PLAN-AMENDMENT-2026-11-10.md`. Runnable instructions: Hua-Sheng's local precise timing handoff. It covers fresh installation, local registration, full numerical replay, paired timing and a separate local PDF; the handoff is not a measurement.

Additional local accepted-precision diagnostics: full-stream precision TODO. This separate diagnostic cannot satisfy the formal 1.05 runtime gate.

Formal main-process timing uses a 2 s calibration target and the frozen 0.95 s minimum observed batch duration. Each batch traverses all 100 saved points. All valid attempts remain visible; a qualified poor result cannot be replaced by a favorable retry. These limits are distinct from the separate traced-cohort diagnostic duration rule.

Main process	Min. baseline s	Min. candidate s	Retained paired attempts
$u\bar{u} \to Z$	—	—	0 · 0 short
$gg \to t\bar{t}$	—	—	0 · 0 short
$u\bar{u} \to Zgg$	—	—	0 · 0 short
$u\bar{u} \to ZZgg$	—	—	0 · 0 short
$u\bar{d} \to W^+$	—	—	0 · 0 short
$e^+e^- \to \mu^+\mu^-$	—	—	0 · 0 short
$e^+e^- \to t\bar{t}\gamma$	—	—	0 · 0 short
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	—	—	0 · 0 short

Minima include all successfully reparsed retained timing attempts and comparison labels, including ineligible short attempts. Missing or invalid observations do not supply a duration. Selected runs and each attempt's exact minima, labels and eligibility remain in the dossier.

8.1 Controls in the formal campaign

Mixed QCD/QED and loop-induced controls require numerical qualification. Their speed measurements are descriptive and do not carry the main-process ≤1.05 runtime gate. Generic raw-summary failure labels can include slow controls; the role-specific interpretation below preserves those labels without turning control speed into a main acceptance failure.

Control	Formal paired numerical evidence	Raw summary status	Runtime gate
$u\bar{u} \to g\gamma$	**pending**	pending	not applicable
$gg \to H$	**pending**	pending	not applicable
$gg \to HH$	**pending**	pending	not applicable

The untimed numerical and provisional shared-host tables are separate datasets. Their completed observations do not populate these formal-campaign fields.

9 Native-DP and rescued cohort diagnostics

Traced full-amplitude subset diagnostics. DP includes internal method selection; compensated DP and naturally rescued QP are separate. Per-call tracing adds overhead. Different frozen subsets cannot be used as backend speed ratios. The diagnostic 90% duration rule is distinct from the main paired 95% rule.

Process	CutTools	Ninja	COLLIER	NinjaRS	TIRust
$u\bar{u} \to Z$	pending	pending	pending	pending	pending
$gg \to t\bar{t}$	pending	pending	pending	pending	pending
$u\bar{u} \to Zgg$	pending	pending	pending	pending	pending
$u\bar{u} \to ZZgg$	pending	pending	pending	pending	pending
$u\bar{d} \to W^+$	pending	pending	pending	pending	pending
$e^+e^- \to \mu^+\mu^-$	pending	pending	pending	pending	pending
$e^+e^- \to t\bar{t}\gamma$	pending	pending	pending	pending	pending
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	pending	pending	pending	pending	pending
$u\bar{u} \to g\gamma$	pending	pending	pending	pending	pending
$gg \to H$	pending	pending	pending	pending	pending
$gg \to HH$	pending	pending	pending	pending	pending

Cells give frozen point counts and diagnostic median µs per call. Empty cohorts have no timing. Short batches, migrations or failure flags cannot yield a valid median. The complete original stream is warmed before membership is frozen; QP timings include failed DP work and natural rescue.

10 Registered implementation and delivery status

Required criterion	Observed status
all eleven five backend numerical	**passed**
eight main runtime gates	**deferred to Hua-Sheng**
existing backend regressions	**deferred to Hua-Sheng**
Installed ABI / Fortran: recorded test groups	**passed**
installer metadata hardening	**passed**
native dp and rescued cohorts	**pending**
paired campaign overall	**deferred to Hua-Sheng**
Current statistics round trip	**passed**
stress accuracy	**open**
Subprocess/event cache lifecycle	**passed**

Installed package. DSO `eb3d67fdc99b742b` Fortran archive `0c86349a4fd0dadd` actual installer source `059bb9c97919`. This is distinct from the registered exporter anchor and each observed export HEAD.

The public Fortran parity replay contains 65 calls per installation. The raw DP replay contains 90 calls per library, with 80 successes and 10 expected failures. Cache transport rejection is checked against the retained old-library negative control. These checks qualify the recorded interface and tested transport/cache rejection behavior; their elapsed times are not physical-amplitude benchmark samples.

Integration work remains open. Selected standalone runtime, control and diagnostic proofs are recorded below. Adaptive-bridge production promotion remains pending a separately bound receipt. The installer metadata correction has a bound production test receipt. These scoped proofs do not close the full campaign, stress or performance gates.

Subprocess/event cache lifecycle: passed. Four fresh workers, 76 checked native-DP physical calls and 96 separate DP/QP tensor-key requests; exact cache/observer parity. Physical-QP rescues, rank-six, EW, poles-only and runtime remain outside this lifecycle scope.

`experiments/subprocess-cache-lifecycle-current-qualification-2/qualification-summary.json`

Current statistics: passed. One retained invocation passed 10 tests and four compiled synthetic Fortran tally/writer subtests, using current production Python readers. Legacy aliases, TIRust/NinjaRS precision counters and aggregation are covered. This qualifies the identified statistics sources; it neither recovers the historical unbound 92-pass invocation nor establishes amplitude, native-reducer or performance results.

`experiments/statistics-roundtrip-current-1/manifest.json`

Higher-rank terminal Fortran qualification is bound to the completed source, IEEE/trap and fixed-input coefficient checks in `experiments/dp-higher-rank-production/validation-summary.json`. Its compiled checks and CutTools parity do not substitute for the registered five-backend campaign or establish runtime admission frequency.

11 Retained accuracy limitations and diagnostics

The report binds the completed portable inventories and their recorded scope; it does not rerun scientific adjudication. Historical library results and private prototypes do not qualify the current installed library. Ordinary numerical and paired runtime gates remain separate. The historical split-dp-terminal-1 QCD n3 optional target and higher-precision workstream are diagnostic disclosures; neither adds a stress acceptance requirement.

Case	Status	Qualification limit
EW threshold mass transport	**open**	Historical diagnostic retained by split-dp-terminal-1: Historical 49af, private mass-transport prototype. QP squaring of existing MP masses reduces the recorded group-5 Gram discrepancy from 1.018% to 1.60e-20, while both arms still refuse 37 threshold tensor parents. Status 120 and accuracy −1 are unchecked; the TIRust sum is incomplete. Matching successful scalar/rank-one subsets does not qualify the failed tensors, full amplitude or performance. `experiments/ew-qp-mass-prototype/observer-native-design/attempt2/execution-1/geometry-audit/README.md`
QCD soft/collinear prototype	**open**	Historical diagnostic retained by split-dp-terminal-1: Historical 49af, six private replays. Topology correction plus Born evaluation at repaired coordinates gives NinjaRS status 322 and a 1.33e-15 analytic double-pole error; topology-only status 322 still misses 3e-6. Finite/single-pole truth remains unavailable. TIRust retains 51 nonsingular conditioning refusals and failed status 420; its incomplete sums are not amplitudes accepted for physics. `experiments/qcd-topology-rescue-native/RESULTS.md`
Default stability estimator	**open**	Historical split-dp-terminal-1 disclosure, same installed eb3d bridge: its completed replay confirms four inaccurate DP acceptances at the production 1e-3 default (maximum finite symmetric error 7.03e-4) and four QP rescues at 1e-6 (maximum symmetric error 5.97e-16). This synthetic host-helper fixture does not qualify physical amplitudes; the registered policy remains 1e-6. `experiments/default-stability-current-1/README.md`
QCD n3 ordinary target	**open**	Historical diagnostic retained by split-dp-terminal-1: Current eb3d campaign: all 100 points pass 3e-6, while the same 28 finite-term 1e-8 misses against CutTools remain. Retained independent QP comparisons support a CutTools DP reference bias and a smaller Ninja/NinjaRS discrepancy at point 39; a reducer cluster alone is not truth. Those historical fixed-QP results are unchecked status 120, not an automatic-rescue certificate. Original comparisons and acceptance gates are unchanged. `campaigns/split-dp-terminal-1/experiments/qcd-n3-target-triage-1/README.md`
Complex-mass extremes	**open**	Historical diagnostic retained by split-dp-terminal-1: Nineteen moderate points agree; original extreme automatic-policy points remain unqualified. Private corrections are unchecked diagnostics. `experiments/complex-mass/README.md`
QCD higher precision	**diagnostic**	Historical split-dp-terminal-1 disclosure: a separate private all-512-bit replay completed 232 tensor calls with actual runtime mapping and exact ordinary-point parity. Forty-six contracted tensor slots retain discrepancies despite close final projections. The changed precision and admission gate do not qualify the adaptive 256-to-512 library, general physical accuracy or runtime acceptance. `experiments/qcd-topology-rescue-higher-precision-native2/README.md`

The historical c91 qualified runtime failure remains an ancestor result with its original source and interval. New source candidates do not erase it. Ordinary-point agreement and a passing eight-process runtime gate alone cannot declare the full integration goal complete.

12 Additional implementation proofs

Each updated claim below is derived from an explicitly selected package whose index and retained files are hash-bound to this dossier. These separate experiments do not replace the source identity, precision usage or numerical observations of the ordinary campaign.

The experimental bridge is an isolated development build, whose source and evidence are recorded in this public feature branch. It has not replaced the installed TIRust bridge used for the ordinary campaign. The branch is `madloop_tirust` the intended future PR target is `madloop_ninjars`.

12.1 Historical split-dp-terminal-1 QCD n4 target misses

The archived split-dp-terminal-1 report retains 45 distinct target-miss points and its old DP/QP counts. Those observations belong to that historical source. Current 1e-8 target outcomes are derived only from this report's closed current checkpoint.

`history/report-before-invariant-symmetry-qualification/archive-manifest.json`

12.2 Experimental adaptive-precision bridge

Separately observed Fortran runtimes now cover debug: 13 genuine binary128 components, 10 adaptive and 1 legacy tensor calls; release: 13 genuine binary128 components, 10 adaptive and 1 legacy tensor calls. Actual 256/512-bit dispatch, cache and check telemetry are retained, including the deliberate error call. These are standalone tensor/transport proofs. Accepted amplitude precision remains unobserved; full-amplitude accuracy, Born behavior and performance are not qualified. Production promotion remains pending a separate receipt.

`experiments/adaptive-qp-fortran-qualification/runtime1/evidence-index.json; experiments/adaptive-qp-release-fortran-qualification-v2/runtime1/evidence-index.json`

12.3 Born/effective-point consistency

Actual backends 8 and 9 complete 8 ordinary and 16 fixed-QP real-Z controls. The 24 retained logs contain 0 W03 warnings; 48 fixed-QP observations carry unchecked status 120. The requested Born/coordinate views and recorded coefficient comparisons pass within this control scope. Unchecked calls do not establish independent stability. CMS/stress coverage, a full campaign and performance remain unqualified.

`experiments/born-effective-point-native89-qualification-1/evidence-index.json`

12.4 Symmetric invariants: component proof

The v2 extracted components pass 5376 DP/QP pairs and 48 trap pairs; 2 deliberate mutations are detected. Caller words, IEEE state, halting and rounding safeguards are retained. This component proof alone does not qualify a physical amplitude.

`experiments/ew-invariant-symmetry-prototype/v2/fortran-fixtures/qualification-1/index.json`

12.5 Symmetric invariants: separate EW stream

The v2 candidate accepts 100 DP / 0 QP points on one frozen 100-point EW stream, versus 82 DP / 18 QP for its old-source comparator. All 6 retained references pass 3e-6 and 1e-8; The retained console has zero W03 lines, and this instrumented diagnostic records zero IEEE state differences. Muted ordinary final-traversal diagnostics are not inferred from those console counts. These observations belong to the separate v2 candidate. They do not replace any of the older campaign's 66 numerical lanes or its provisional timing identity, and supply no runtime acceptance.

`experiments/ew-invariant-symmetry-prototype/v2/runtime-1/qualification-1/index.json`

12.6 Positive-width CMS diagnostic remains open

The 15 new positive-width rows retain 6 DP refusals, 6 returned QP tensors and 3 returned COLLIER tensors. All 15 rows retain IEEE flags; the combined analysis preserves 191 open items, including ancestral controls. Protocol completion is not CMS acceptance.

`experiments/generated-rank6-poles-proposal/attempt1/cms-width1/runtime1/index.json`

12.7 Common-input tensor diagnostic remains open

All 30 synthetic rows have exact effective inputs. The retained comparison still reports 3 comparison discrepancies, 6 refusals and 18 host IEEE observations. These are retained comparator observations, not an attribution of fault to a reducer. No physical or performance qualification follows.

`experiments/generated-rank6-poles-proposal/attempt1/common-input-proposal/runtime1/index.json`

12.8 Fixed-QP mass transport remains open

Nonmass input comparisons pass in the retained transport diagnostic. TIRust refusal counts by its two saved points are [0, 37] for the original arm and [0, 37] for the candidate arm. QP exception observations remain retained. Threshold aggregates are incomplete; checked-QP, physical accuracy and performance remain unqualified.

`experiments/ew-qp-mass-prototype/observer-native-design/attempt2/execution-1/evidence-index.json`

12.9 Installer metadata hardening

The strict JSON version-type correction is applied to production installer source 4397d9f60e146354. The bound receipt records 27 passing tests and 37 passing subtests, with the existing qualified prefix still available and QP-capable. Receipt provenance: Transcribed from actual completed tools session52323; no raw pytest log captured. This closes the selected metadata-hardening item; it does not supply new native or numerical qualification.

`experiments/installer-metadata-promotion-1/index.json`

12.10 Symmetric invariants: Fortran template promoted

The applied Fortran template is 4ccd4456d0b28f39. Its bound source-comparison receipt records 316 matching canonical statements in the qualified EW adapter context. All 4 tests pass: 3 compiled fixture methods and 1 Python permutation method. No real reducer or new physical replay is invoked by these tests. New exports can use this template; the older campaign's 66 lanes and provisional timings remain tied to their original source. Adaptive-bridge production promotion and final performance acceptance remain separate.

`experiments/ew-invariant-symmetry-prototype/v2/promotion-qualification-1/index.json`

12.11 Higher-rank finite residual clarification

The additive audit separates the worst finite symmetric residual, approximately 0.001328503784, from the aggregate maximum 1 in the simple-pole slot. The aggregate value includes near-zero UV comparison noise; it is not the finite residual. The original diagnostic remains preserved. Finite discrepancies and the comparator's requested-accuracy observations remain open; this offline audit establishes neither a reducer fault nor physical accuracy or performance.

`experiments/generated-rank6-poles-proposal/attempt1/common-input-proposal/finite-audit-1/index.json`

12.12 Adaptive fixed-QP private replay

The completed private replay serves 400 calls, recovers 37 former refusals, preserves 363 formerly served outputs bitwise, and changes W03 lines from 74 to zero. Both public statuses remain unchecked fixed-mode 120. The additive offline classification correction retains the original controller's complete=false failure and makes no native retry. Successful service establishes neither independent truth nor checked stability or per-call 512-bit execution. This is retained synthetic/private diagnostic evidence, separate from the current campaign's installed bridge. Physical amplitudes, general stress, production promotion and performance remain unqualified; recorded wall durations are administrative.

`experiments/adaptive-qp-physical-replay-qualification-1/evidence-index.json`

12.13 Adaptive replay: independent backend8 audit

Backend8 final sums agree with the private adaptive replay at maximum symmetric residuals at most 2.685e-16 (ordinary) and 2.029e-16 (threshold). Both serve 400 calls with the same physical/Born inputs. All 200 threshold native scales differ: backend8 squares in DP before promotion, backend9 after promotion. This is not identical native input data or checked truth; fixed-mode status120 remains unchecked. This is retained synthetic/private diagnostic evidence, separate from the current campaign's installed bridge. Physical amplitudes, general stress, production promotion and performance remain unqualified; recorded wall durations are administrative.

`experiments/adaptive-qp-physical-replay-independent-8-audit-1/evidence-index.json`

12.14 Adaptive masked checks and retained trapped failure

Masked adaptive threshold output passes independent automatic checks with status323: 600 QP calls served, 489 former served outputs preserved bitwise and 111 former refusals recovered across observed modes [4,5,4]. Ordinary status219 is accepted in DP and adds no QP coverage. Both original and adaptive trapped workers SIGFPE during threshold DP warmup2; the combined trap gate remains false. Offline evidence localizes the shared DP Cayley DIVSD instruction; the precise hardware exception remains inferred. Explicit rotation identity and per-call precision remain unavailable. This is retained synthetic/private diagnostic evidence, separate from the current campaign's installed bridge. Physical amplitudes, general stress, production promotion and performance remain unqualified; recorded wall durations are administrative.

`experiments/adaptive-qp-checked-trap-proposal-1/qualification-1/evidence-index.json`

12.15 Rank-six Feynman parameters: all-component fixture cross-check

The later independent integration corroborates all 210 finite components (420 parts) of the same synthetic fixture: 420 point-estimate passes at 3e-6 and 1e-8, maximum TIRust residual at most 9.625e-20. COLLIER passes 298/420 at 3e-6 and 92/420 at 1e-8; original answers and flag -1 remain retained. Analytic error alone preserves 420 TIRust passes at 3e-6; at 1e-8 it gives 357 agreements and 63 inconclusive parts. The vector arithmetic has a 64-bit significand. Analytic bounds exclude finite node/weight/arithmetic error; convergence supports a practical cross-check, without a whole-program certificate or cache-bug attribution. This is retained synthetic/private diagnostic evidence, separate from the current campaign's installed bridge. Physical amplitudes, general stress, production promotion and performance remain unqualified; recorded wall durations are administrative.

`experiments/rank6-feynman-parameter-all-components-qualification-1/index.json`

12.16 Rank-six Feynman parameters: seven-component cross-check

Independent direct integration corroborates 7 sampled finite components (14 real/imaginary parts); maximum observed TIRust residual at most 4.762e-17. Five of six disputed COLLIER parts miss 3e-6, consistent with retained flag -1. The analytic-error-only maximum upper residual is at most 3.558e-7. Seven samples do not qualify all 210 components. Finite node/weight/arithmetic error is checked empirically; no whole-program certificate or COLLIER cache-bug attribution is supplied. This is retained synthetic/private diagnostic evidence, separate from the current campaign's installed bridge. Physical amplitudes, general stress, production promotion and performance remain unqualified; recorded wall durations are administrative.

`experiments/rank6-feynman-parameter-qualification-1/index.json`

12.17 Guarded adaptive bridge: closed private compilation

The guarded private release build completes with four newly compiled crates and 140 reused dependency artifacts. All 1,193 protected bindings remain unchanged; 499 source identities and 133 retained code texts bind the sole reviewed signed-zero denominator guard. The new library 69722452c234f494 is distinct from the retained old private c5de24e6c41a90ca library and the campaign's eb3d67fdc99b742b bridge. Both prior failed builds remain retained: LLVM allocation failure under a 4 GiB address-space limit, then a separate 600-second deadline. The Cargo-banner verifier correction is retained without another build or DSO read. This package qualifies compilation and source provenance only. The new library was not loaded, installed or numerically exercised in this build package; physical accuracy, precision, stability, trap behavior, production promotion and runtime performance remain unqualified here. Recorded build durations are administrative and supply no performance claim.

`experiments/cayley-guarded-adaptive-build-qualification-1/evidence-index.json`

12.18 Guarded adaptive physical replay: masked equality, new trapped failures

The guarded private bridge preserves both closed adaptive masked references bitwise. Checked mode accepts the ordinary point in DP (status219) and the threshold point in QP (status323), with 600 QP calls in modes [4,5,4]. Forced masked QP serves 400 calls; status120 and accuracy -1 remain unchecked. The original-backend comparison remains a nonreference diagnostic: its permissive regression passes despite maximum relative residual 0.9241097961, nonidentical outputs and the retained threshold status420 with 111 refusals. Both guarded trapped workers SIGFPE before observer journals; their FileNotFoundError assessments remain retained. All joint numerical, host-stability, QP-coverage and trap gates remain false. Offline localization identifies other DIVSD sites in DP gram_indicator and multiprecision dilogarithm series; the actual operands and exact exception subtype remain unknown. The 69722452 guarded library, old c5de24e6

private library and campaign eb3d67fd bridge remain distinct. Two optional verifier-test policy failures are retained separately from the passing portable replay and physical outcomes. This two-point private diagnostic establishes neither independent physical truth, explicit rotation identity, per-call working precision, production promotion nor runtime performance. Recorded execution duration is administrative; precise timing remains a local TODO.

`experiments/cayley-guarded-adaptive-physical-qualification-1/evidence-index.json`

12.19 Current CMS coverage

incomplete: 6 expected automatic requests. This selected closed capsule records 6 amplitude streams started; 6 completed. Each CMS final stream requires 21 points and 42 order outcomes; warmup, bootstrap and final phases remain separate. Integrity/preparation/preflight do not establish checked full-stream agreement. Failed, unexecuted and unchecked outcomes remain retained. No repaired-geometry EW oracle, broad stress/CMS acceptance or performance qualification follows. Native orchestration completed 6 requests, while 0 complete streams were host checked and 252 final order outcomes remain unexecuted. Individual recorded warmup acceptance does not qualify a request or imply READY/RUN/final traversal. Worker launches, observed amplitude entry, bootstrap rows and repeated warmup observations do not determine the hidden amplitude evaluation count. Preparation/preflight are inherited from explicitly indexed prior setup; this capsule does not claim fresh CLI exports or builds. Explicit failed ancestor capsules and all their request rows remain retained separately; successor observations do not erase them.

`experiments/current-cms-replay-retention-2/evidence-index.json`

12.20 Current physical stress coverage

failed: 24 expected automatic requests. This selected closed capsule records 4 amplitude streams started; 4 completed. Each stress stream has two saved points. Integrity/preparation/preflight do not establish checked full-stream agreement. Failed, unexecuted and unchecked outcomes remain retained. No repaired-geometry EW oracle, broad stress/CMS acceptance or performance qualification follows. 4 host-checked two-point streams supply 8 qualified final point evaluations and 12 observed warmup calls; these observations do not count hidden evaluations. 5 of 6 checked-subset comparisons fail the 3e-6 criterion. Complete six-lane coverage and agreement remain separate gates. The retained comparator's ordinary_target_passed field applies the 1e-8 target to the whole two-point stream. Deferred and not-started requests remain explicit; the stage outcome is not replaced by a passing subset. The stage ended with a lock timeout before backend 8 launched. Four threshold streams were host checked; five of their six pairwise comparisons fail the regression criterion. One stream was deferred and 19 were not started. Preparation/preflight are inherited from explicitly indexed prior setup; this capsule does not claim fresh CLI exports or builds.

`experiments/physical-stress-current-portable-4-terminal-2/evidence-index.json`

12.21 Closed current QCD n3 target investigation

The bounded offline investigation retains the same 1,200 rows and 4,800 coefficient values as the prior closed result, including the same 28 finite-only 1e-8 target-miss points. All Born and pole comparisons pass 1e-8. The largest CutTools/TIRust symmetric finite residual is approximately 2.806420647e-6 at point 93; the largest non-CutTools pair residual is approximately 8.595156396e-8 at point 39. Cross-reducer consistency supplies no independent truth claim. This seven-input investigation runs no new native or QP evaluation and cannot replace the main full11 checkpoint or change any qualification or performance gate.

`experiments/invariant-symmetry-qcd-n3-target-analysis/portable-analysis2/index.json`

Actual Fortran tensor transport, unchecked fixed-QP controls and a separate physical stream each have a defined scope. They do not establish adaptive full-amplitude accuracy, general stress qualification or final runtime acceptance. The older campaign remains unchanged.

13 Retrospective SM stability · $gg \to t\bar{t}$

Accepted composite host diagnostic after rescue

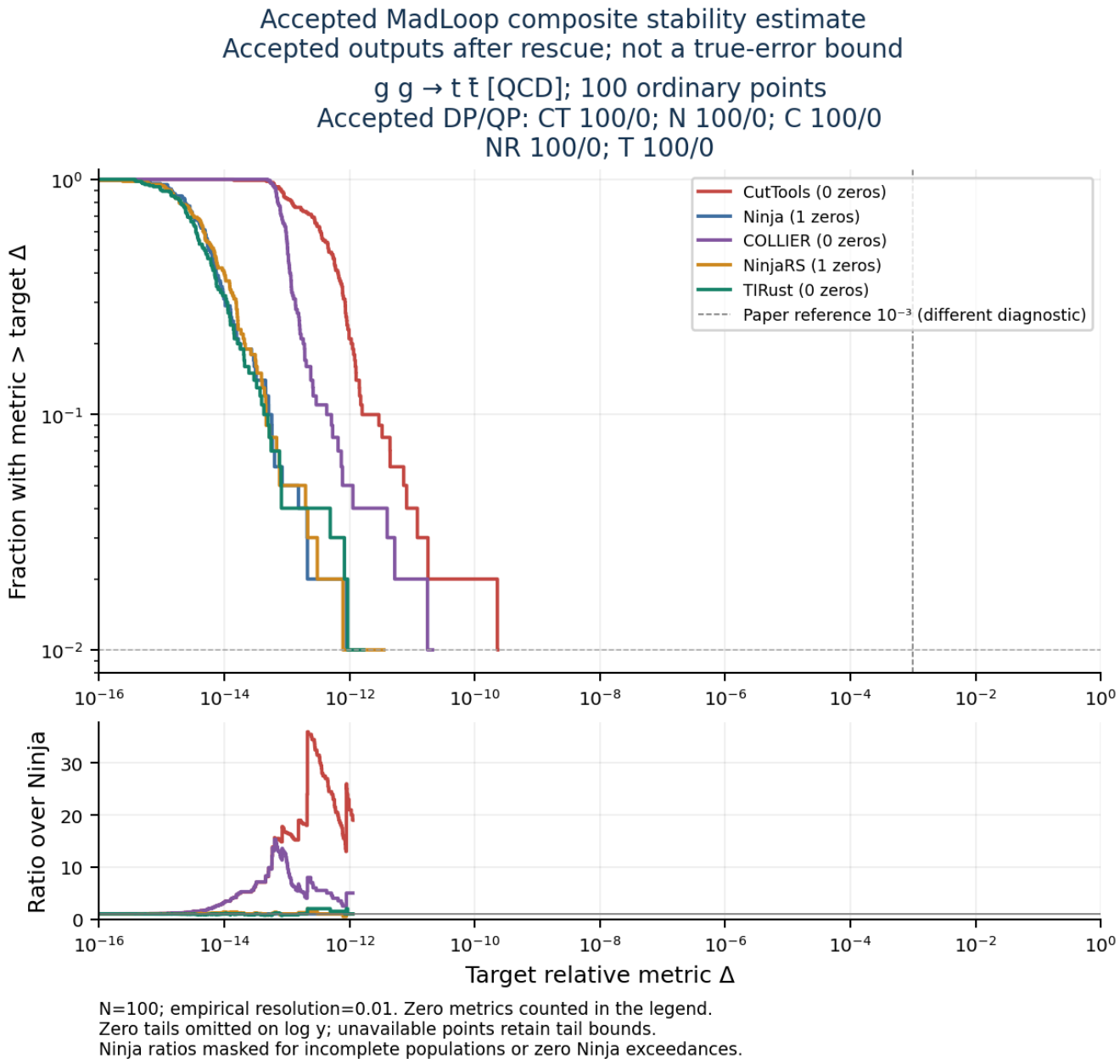


Figure 1: Recorded post-rescue MadLoop composite estimates include finite and pole information. Estimator definitions differ, including COLLIER's; these curves do not rank true numerical accuracy. Exact-zero estimates are recorded zero diagnostics, not certified digits or infinite precision. Accepted DP/QP counts describe final outcomes, not all internal attempts. The 10^{-3} line is a paper reference for a different diagnostic, not an independently byte-bound host-policy marker. Each curve uses the same 100 ordinary saved points, with 1% empirical tail resolution. These are post-rescue accepted-output diagnostics, not raw-DP probes or asserted paper-cut-qualified fixtures. Zero tails are omitted on the log axis; ratios over Ninja are masked for a zero denominator, without a floor or pseudocount. No rarer population-tail, independent-truth or performance acceptance follows.

Indexed PDF export: `stability_appendix_c/retrospective-main-closed11-1/figures/qcd_n2_gg_ttx-accepted-internal-estimate.pdf`

14 Retrospective SM stability · $gg \to t\bar{t}$

Finite disagreement with current candidate CutTools

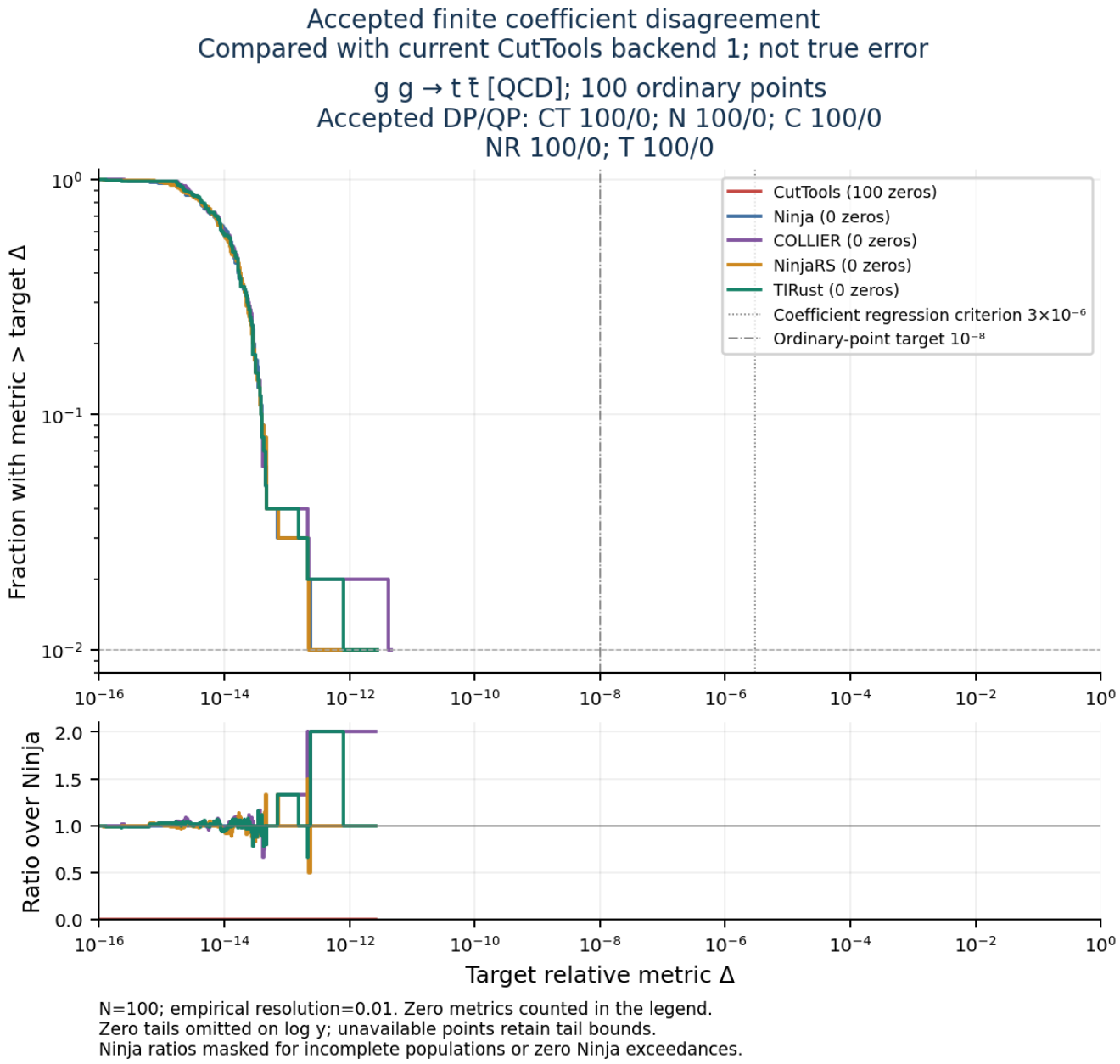


Figure 2: The finite metric is `abs(a-b)/abs(a+b)` against current candidate CutTools. That reference is not independently known truth; its 100 self-comparisons are exactly zero. The 3e-6 regression reference and stricter 1e-8 target remain distinct, and retained target misses are not discarded. Agreement with this reference does not rank true accuracy. Each curve uses the same 100 ordinary saved points, with 1% empirical tail resolution. These are post-rescue accepted-output diagnostics, not raw-DP probes or asserted paper-cut-qualified fixtures. Zero tails are omitted on the log axis; ratios over Ninja are masked for a zero denominator, without a floor or pseudocount. No rarer population-tail, independent-truth or performance acceptance follows.

Indexed PDF export: `stability_appendix_c/retrospective-main-closed11-1/figures/qcd_n2_gg_ttx-finite-disagreement-current-cuttools.pdf`

15 Retrospective SM stability · $u\bar{u} \to Zgg$

Accepted composite host diagnostic after rescue

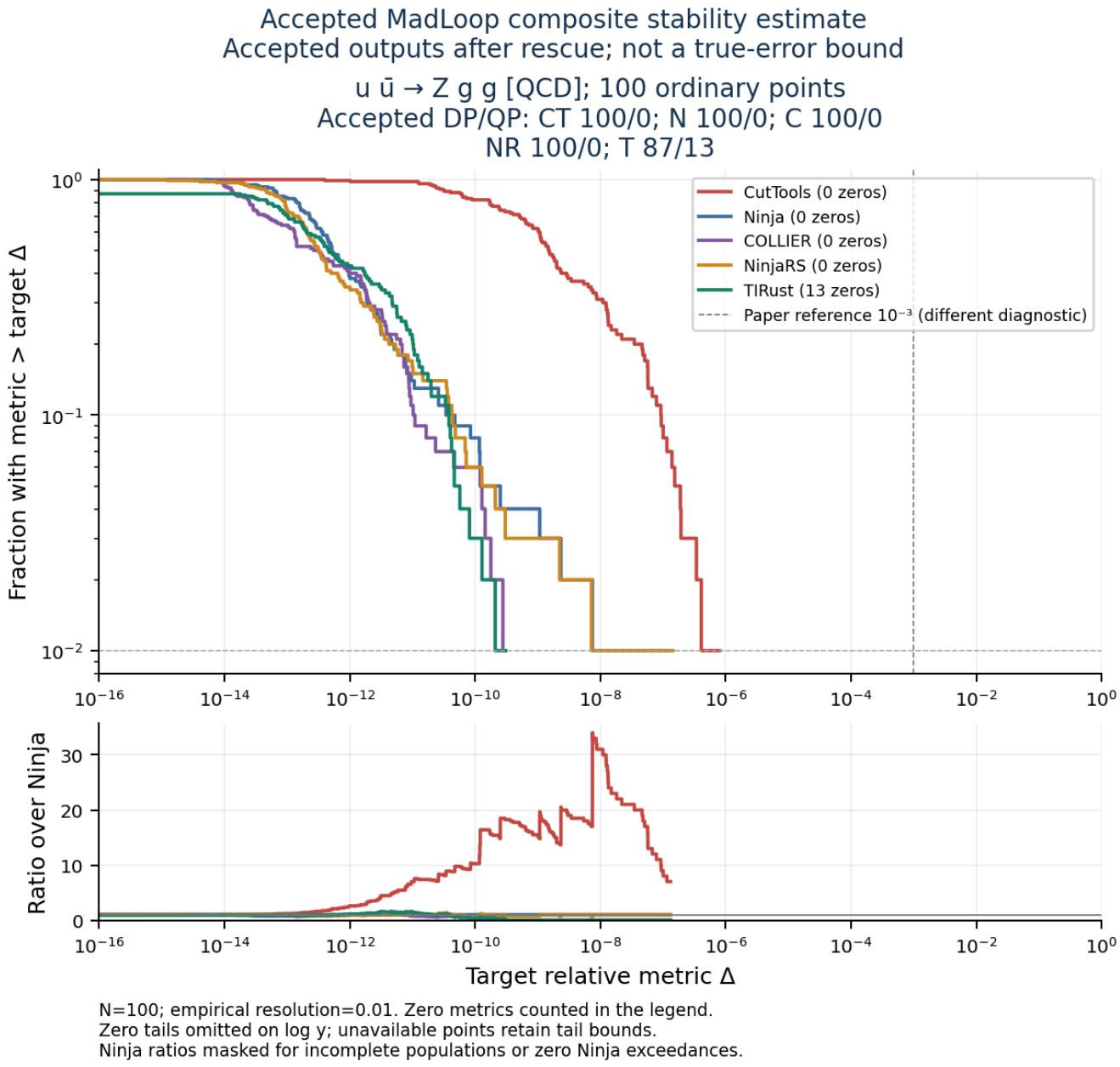


Figure 3: Recorded post-rescue MadLoop composite estimates include finite and pole information. Estimator definitions differ, including COLLIER's; these curves do not rank true numerical accuracy. Exact-zero estimates are recorded zero diagnostics, not certified digits or infinite precision. Accepted DP/QP counts describe final outcomes, not all internal attempts. The 10^{-3} line is a paper reference for a different diagnostic, not an independently byte-bound host-policy marker. Each curve uses the same 100 ordinary saved points, with 1% empirical tail resolution. These are post-rescue accepted-output diagnostics, not raw-DP probes or asserted paper-cut-qualified fixtures. Zero tails are omitted on the log axis; ratios over Ninja are masked for a zero denominator, without a floor or pseudocount. No rarer population-tail, independent-truth or performance acceptance follows.

Indexed PDF export: `stability_appendix_c/retrospective-main-closed11-1/figures/qcd_n3_uux_zgg-accepted-internal-estimate.pdf`

16 Retrospective SM stability · $u\bar{u} \to Zgg$

Finite disagreement with current candidate CutTools

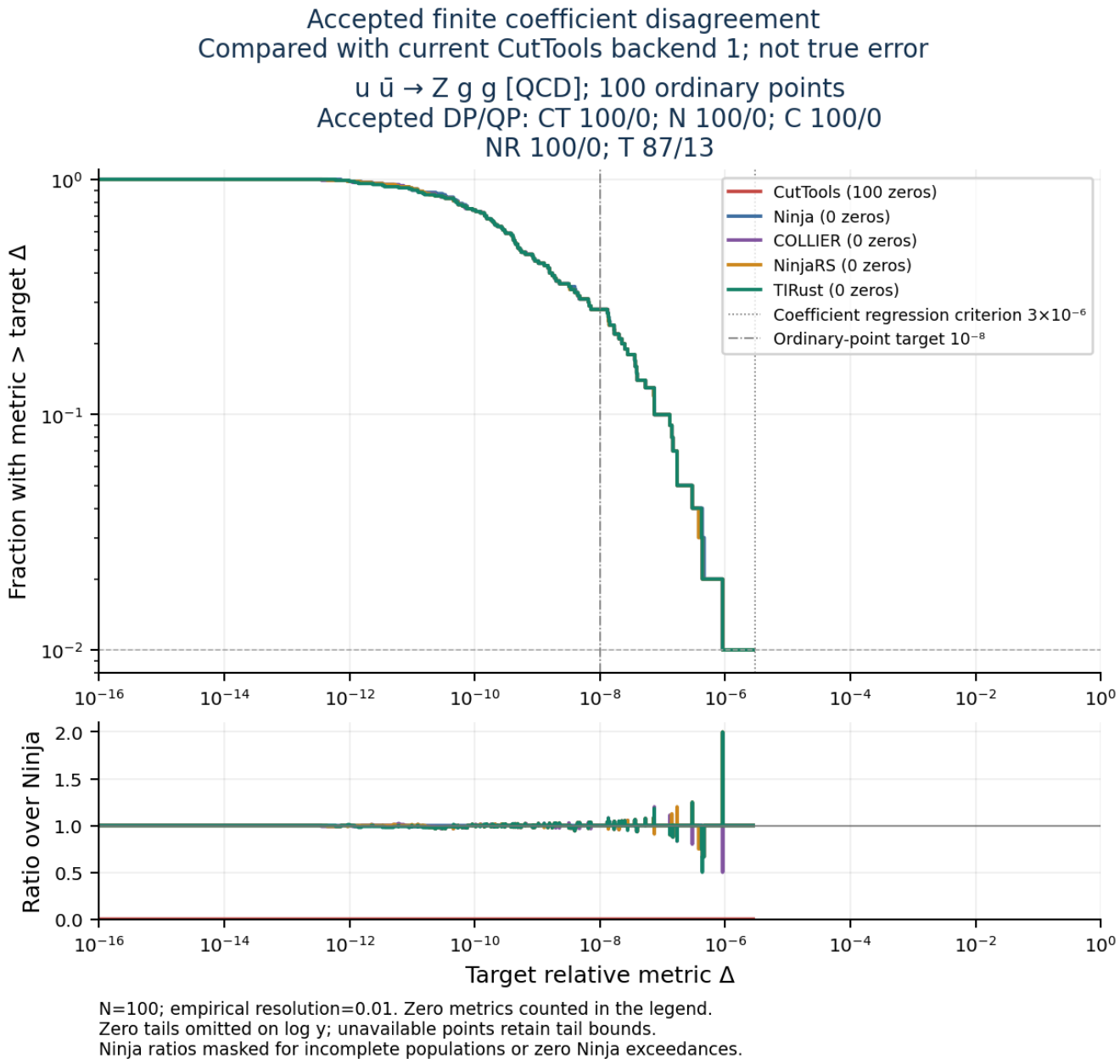


Figure 4: The finite metric is `abs(a-b)/abs(a+b)` against current candidate CutTools. That reference is not independently known truth; its 100 self-comparisons are exactly zero. The 3e-6 regression reference and stricter 1e-8 target remain distinct, and retained target misses are not discarded. Agreement with this reference does not rank true accuracy. Each curve uses the same 100 ordinary saved points, with 1% empirical tail resolution. These are post-rescue accepted-output diagnostics, not raw-DP probes or asserted paper-cut-qualified fixtures. Zero tails are omitted on the log axis; ratios over Ninja are masked for a zero denominator, without a floor or pseudocount. No rarer population-tail, independent-truth or performance acceptance follows.

Indexed PDF export: `stability_appendix_c/retrospective-main-closed11-1/figures/qcd_n3_uux_zgg-finite-disagreement-current-cuttools.pdf`

17 Retrospective SM stability · $e^+e^- \to \mu^+\mu^-$

Accepted composite host diagnostic after rescue

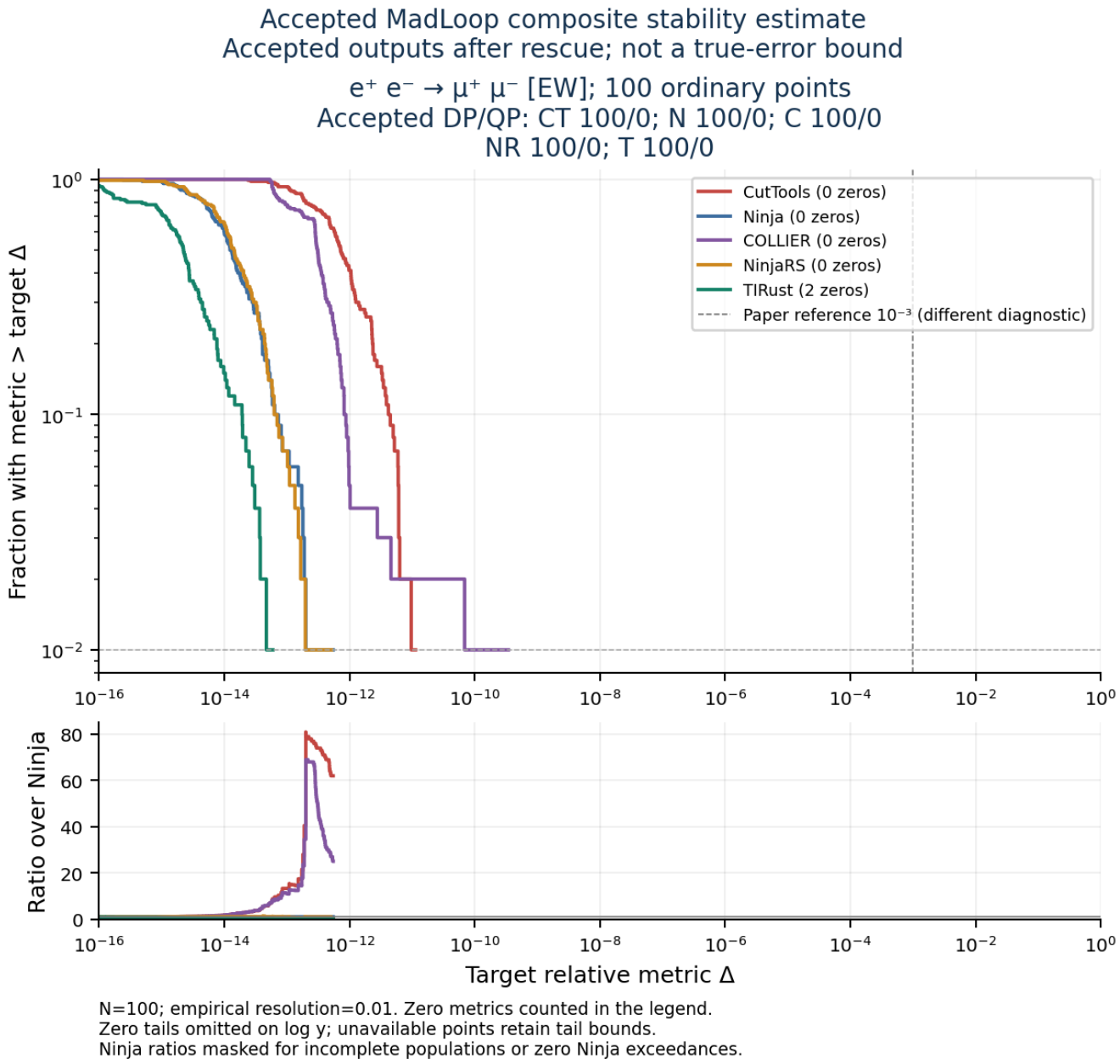


Figure 5: Recorded post-rescue MadLoop composite estimates include finite and pole information. Estimator definitions differ, including COLLIER's; these curves do not rank true numerical accuracy. Exact-zero estimates are recorded zero diagnostics, not certified digits or infinite precision. Accepted DP/QP counts describe final outcomes, not all internal attempts. The 10^{-3} line is a paper reference for a different diagnostic, not an independently byte-bound host-policy marker. Each curve uses the same 100 ordinary saved points, with 1% empirical tail resolution. These are post-rescue accepted-output diagnostics, not raw-DP probes or asserted paper-cut-qualified fixtures. Zero tails are omitted on the log axis; ratios over Ninja are masked for a zero denominator, without a floor or pseudocount. No rarer population-tail, independent-truth or performance acceptance follows.

Indexed PDF export: `stability_appendix_c/retrospective-main-closed11-1/figures/ew_n2_epem_mupmum-accepted-internal-estimate.pdf`

18 Retrospective SM stability · $e^+e^- \to \mu^+\mu^-$

Finite disagreement with current candidate CutTools

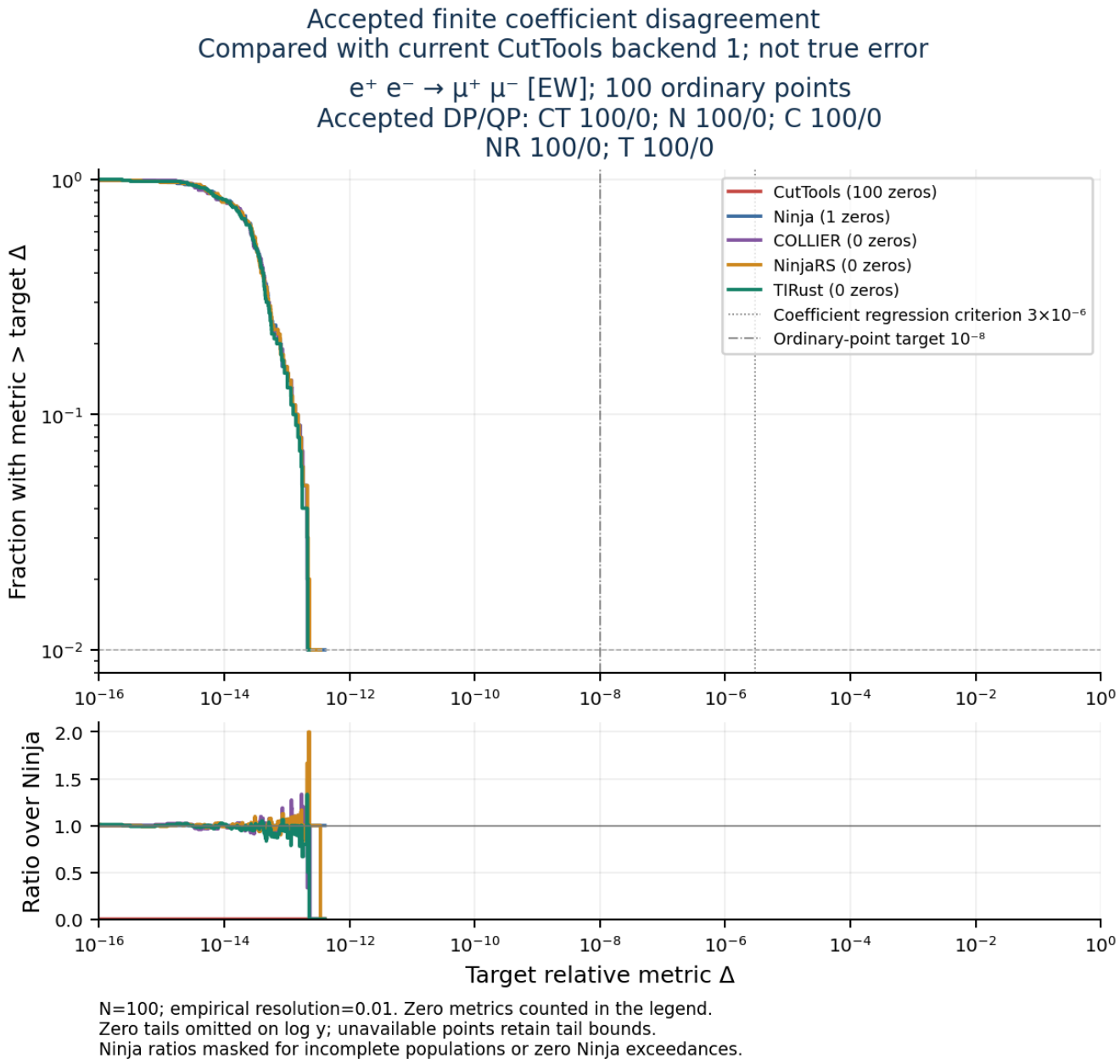


Figure 6: The finite metric is `abs(a-b)/abs(a+b)` against current candidate CutTools. That reference is not independently known truth; its 100 self-comparisons are exactly zero. The 3e-6 regression reference and stricter 1e-8 target remain distinct, and retained target misses are not discarded. Agreement with this reference does not rank true accuracy. Each curve uses the same 100 ordinary saved points, with 1% empirical tail resolution. These are post-rescue accepted-output diagnostics, not raw-DP probes or asserted paper-cut-qualified fixtures. Zero tails are omitted on the log axis; ratios over Ninja are masked for a zero denominator, without a floor or pseudocount. No rarer population-tail, independent-truth or performance acceptance follows.

Indexed PDF export: `stability_appendix_c/retrospective-main-closed11-1/figures/ew_n2_epem_mupmum-finite-disagreement-current-cuttools.pdf`

19 Retrospective SM stability · $u\bar{u} \to ZZgg$

Accepted composite host diagnostic after rescue

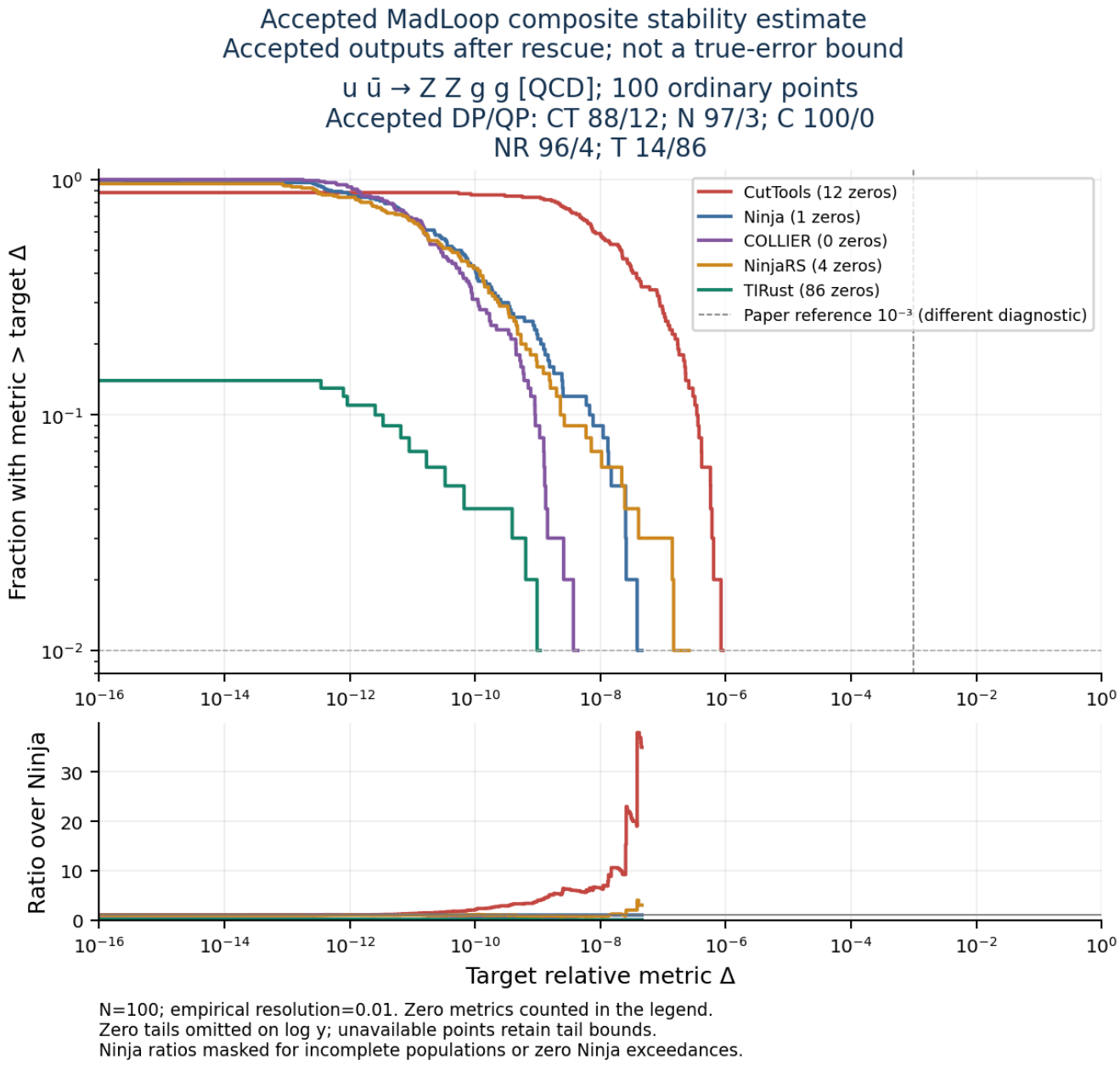


Figure 7: Recorded post-rescue MadLoop composite estimates include finite and pole information. Estimator definitions differ, including COLLIER's; these curves do not rank true numerical accuracy. Exact-zero estimates are recorded zero diagnostics, not certified digits or infinite precision. TIRust has 86 zero estimates alongside 14 DP / 86 QP final acceptances; most of these zeros are associated with accepted QP diagnostics. Accepted DP/QP counts describe final outcomes, not all internal attempts. The 10^{-3} line is a paper reference for a different diagnostic, not an independently byte-bound host-policy marker. Each curve uses the same 100 ordinary saved points, with 1% empirical tail resolution. These are post-rescue accepted-output diagnostics, not raw-DP probes or asserted paper-cut-qualified fixtures. Zero tails are omitted on the log axis; ratios over Ninja are masked for a zero denominator, without a floor or pseudocount. No rarer population-tail, independent-truth or performance acceptance follows.

Indexed PDF export: `stability_appendix_c/retrospective-main-closed11-1/figures/qcd_n4_uux_zzgg-accepted-internal-estimate.pdf`

20 Retrospective SM stability · $u\bar{u} \to ZZgg$

Finite disagreement with current candidate CutTools

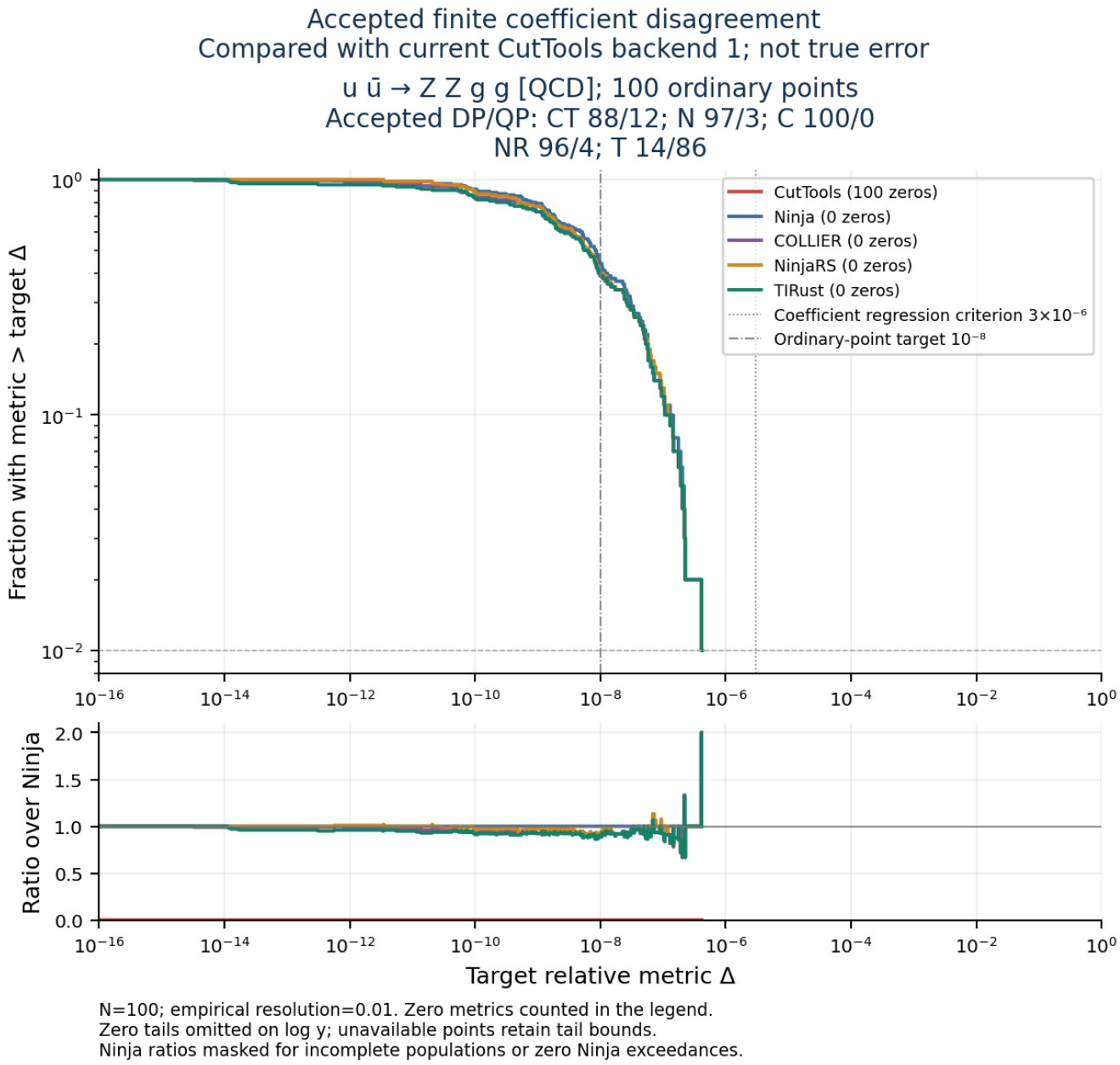


Figure 8: The finite metric is `abs(a-b)/abs(a+b)` against current candidate CutTools. That reference is not independently known truth; its 100 self-comparisons are exactly zero. The 3e-6 regression reference and stricter 1e-8 target remain distinct, and retained target misses are not discarded. Agreement with this reference does not rank true accuracy. Each curve uses the same 100 ordinary saved points, with 1% empirical tail resolution. These are post-rescue accepted-output diagnostics, not raw-DP probes or asserted paper-cut-qualified fixtures. Zero tails are omitted on the log axis; ratios over Ninja are masked for a zero denominator, without a floor or pseudocount. No rarer population-tail, independent-truth or performance acceptance follows.

Indexed PDF export: `stability_appendix_c/retrospective-main-closed11-1/figures/qcd_n4_uux_zzgg-finite-disagreement-current-cuttools.pdf`

21 Appendix C style SM stability pilot

These four plots use 100 cut-qualified points per process, with 1% empirical tail resolution. The host forces its DP path and selects each configured backend through the card. Actual reducer identity and reducer-internal arithmetic precision are unobserved; unchecked status 110 is not checked host acceptance. This three-probe internal-consistency diagnostic supplies neither independently known true error nor performance acceptance. The existing retrospective plots have a different scope.

For the three retained probe values, MadLoop's range-over-absolute-mean convention is $\Delta = \frac{\max(f_1,f_2,f_3)-\min(f_1,f_2,f_3)}{\left|\frac{f_1+f_2+f_3}{3}\right|}$. Here f_i is the retained finite coefficient divided by Born. The legacy stability driver's common $\frac{\alpha_s}{2\pi}$ factor cancels algebraically in relative spread; it is omitted from the pilot's f_i. No bitwise equivalence to separately scaled floating-point values is implied. All-zero values give zero; nonzero spread with zero mean gives infinity. The exact formula comes from MadLoop's implementation; the paper describes the probes and tail presentation, rather than printing this equation.

The tail uses strict exceedance $P(\Delta > \delta)$ over all 100 intended point slots. Known exceedances give a lower bound; dotted upper bounds account for unavailable outcomes and are not measured failure fractions. A wholly unavailable lane is unknown in $[0, 1]$; its zero lower bound cannot appear on a log axis. Complete populations have exact empirical tails. Ninja ratios require complete paired populations and a positive Ninja denominator, with no floor or pseudocount. A valid infinite spread remains distinct from an unavailable nonfinite-coefficient failure.

All twenty configured requests and their native failures remain in the dossier and the tables below. Each lane has 100 intended physical points and 300 intended primary probe-call outcomes. Bootstrap calls are separate. A complete request can still have partially unavailable diagnostic points; completion does not establish a measured complete tail. The frozen protocol's broad precision wording does not establish that every internal arithmetic operation was observed in binary64.

The ordinary full-eleven checkpoint, physical-stress and CMS outcomes, precision cohorts and Hua-Sheng's deferred local performance acceptance keep their existing gates. This pilot changes none of them.

22 Raw stability pilot · $gg \to t\bar{t}$

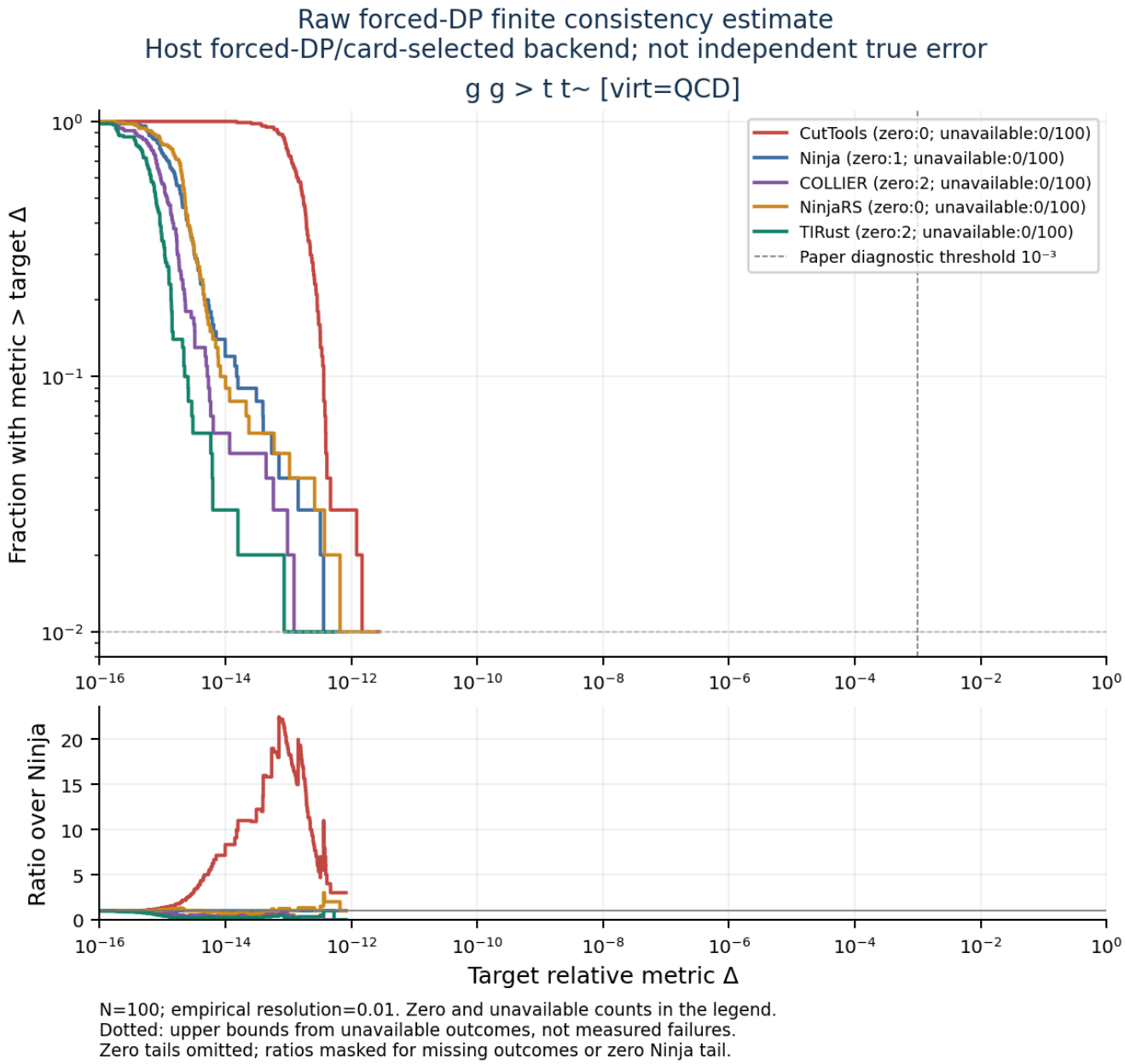


Figure 9: Host forced-DP/card-selected backend; actual backend and internal precision unobserved. Three-probe consistency on 100 cut-qualified points (1% empirical resolution), not true error. Dotted curves are upper bounds for unavailable outcomes, not measured failure fractions. Zero tails are omitted on log axes; missing populations or zero Ninja tails mask ratios. All five configured reducers remain visible. No performance acceptance.

Configured reducer	Result state	Available / unavailable	Zero / infinite	Tail bounds at 1e-3
CutTools	complete	100 / 0	0 / 0	0%–0%
Ninja	complete	100 / 0	1 / 0	0%–0%
COLLIER	complete	100 / 0	2 / 0	0%–0%
NinjaRS	complete	100 / 0	0 / 0	0%–0%
TIRust	complete	100 / 0	2 / 0	0%–0%

Indexed PDF export: `stability_appendix_c/raw-dp-pilot-5-figures-1/figures/qcd_n2_gg_ttx.pdf`

23 Raw stability pilot · $u\bar{u} \to Zgg$

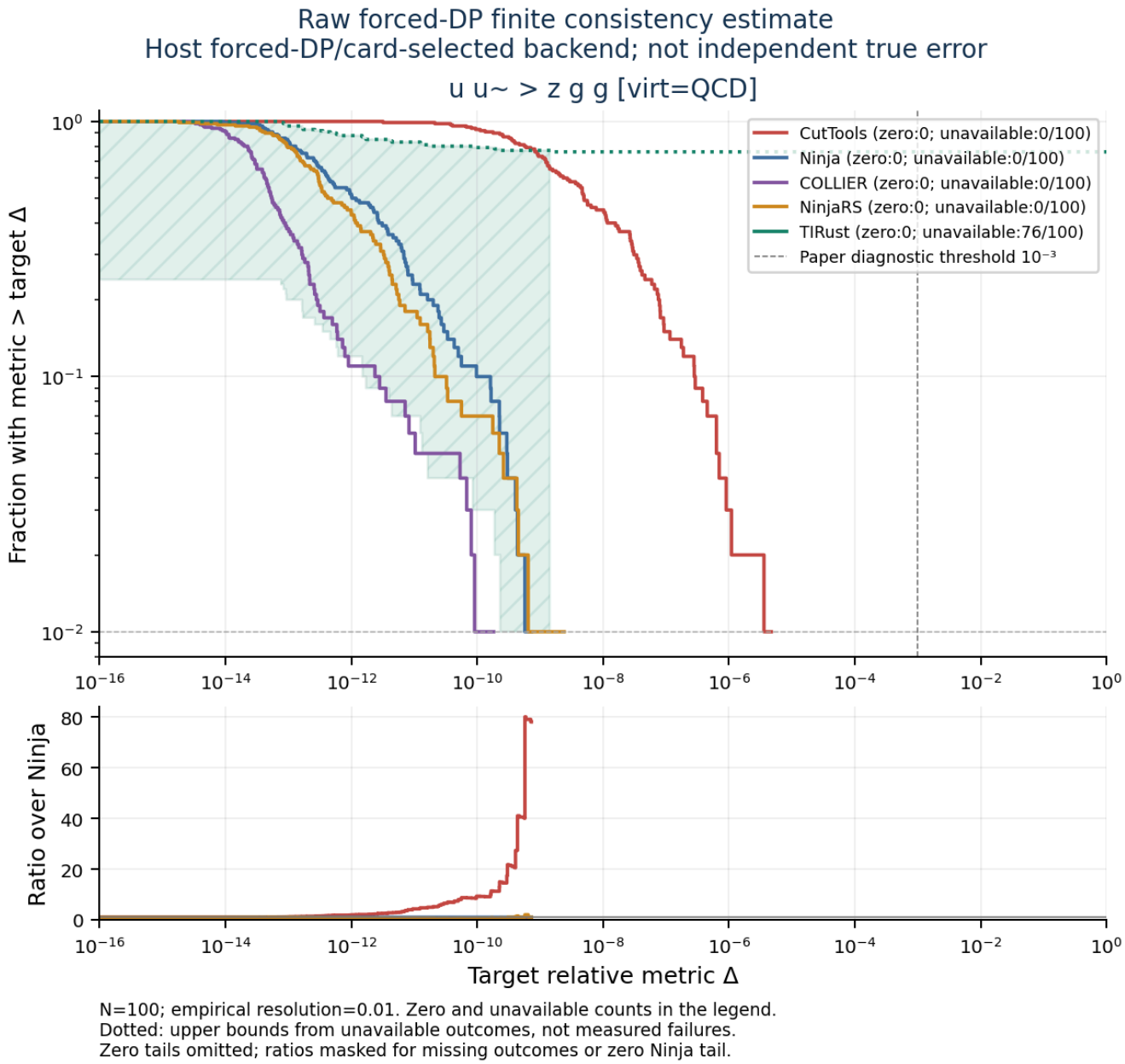


Figure 10: Host forced-DP/card-selected backend; actual backend and internal precision unobserved. Three-probe consistency on 100 cut-qualified points (1% empirical resolution), not true error. Dotted curves are upper bounds for unavailable outcomes, not measured failure fractions. Zero tails are omitted on log axes; missing populations or zero Ninja tails mask ratios. All five configured reducers remain visible. No performance acceptance.

Configured reducer	Result state	Available / unavailable	Zero / infinite	Tail bounds at 1e-3
CutTools	complete	100 / 0	0 / 0	0%–0%
Ninja	complete	100 / 0	0 / 0	0%–0%
COLLIER	complete	100 / 0	0 / 0	0%–0%
NinjaRS	complete	100 / 0	0 / 0	0%–0%
TIRust	complete	24 / 76	0 / 0	0%–76%

TIRust's request completed, but only its first 24 point diagnostics are available. The following 76 are unavailable after the point-25 group failure and persistent DP floating-point latch. Its 1e-3 tail is bounded by 0%–76%, not a measured zero tail. The 15 active-group failure calls and 228 latched calls do not mean 76 independent exceptions.

Indexed PDF export: `stability_appendix_c/raw-dp-pilot-5-figures-1/figures/qcd_n3_uux_zgg.pdf`

24 Raw stability pilot · $e^+e^- \to \mu^+\mu^-$

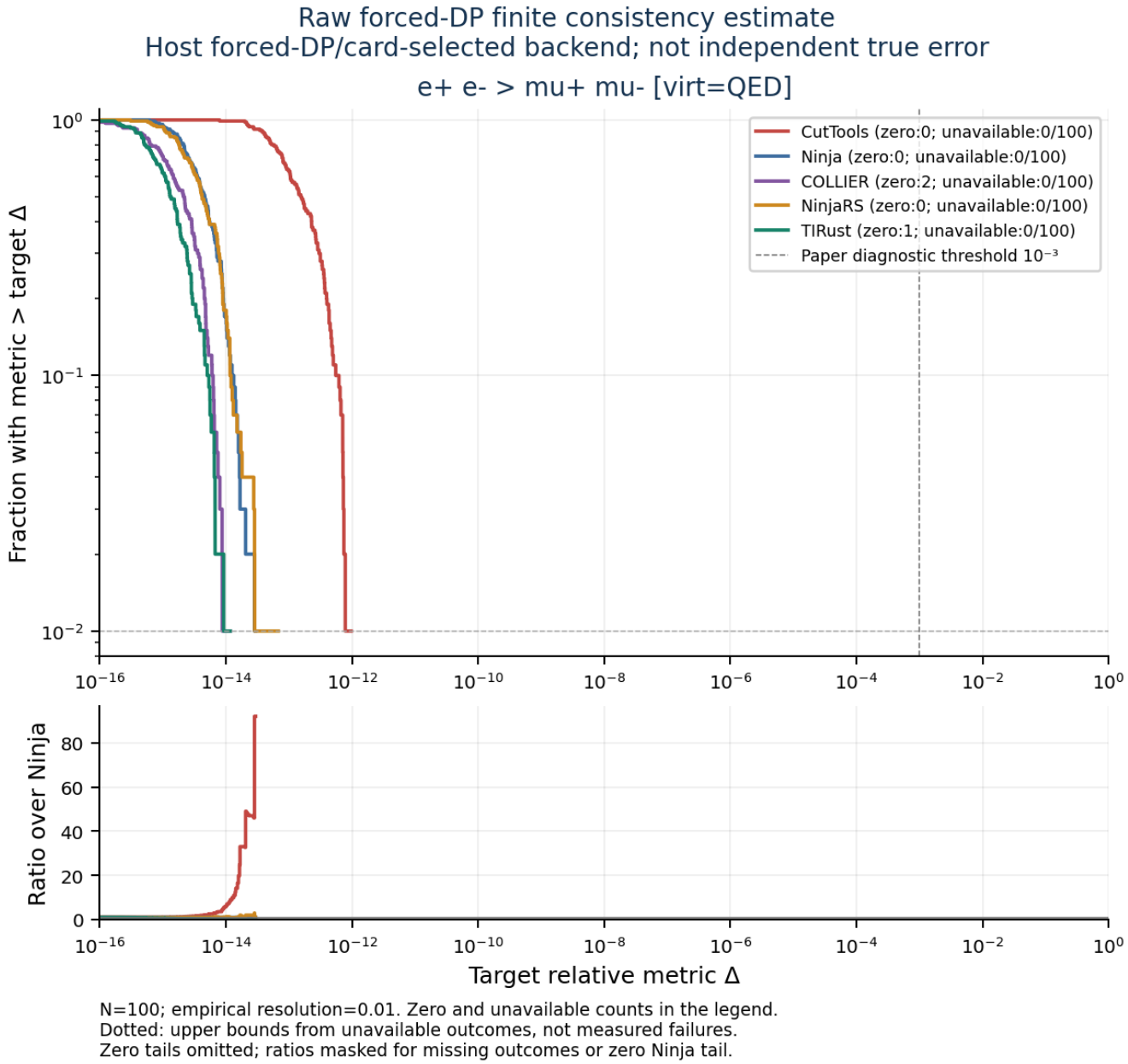


Figure 11: Host forced-DP/card-selected backend; actual backend and internal precision unobserved. Three-probe consistency on 100 cut-qualified points (1% empirical resolution), not true error. Dotted curves are upper bounds for unavailable outcomes, not measured failure fractions. Zero tails are omitted on log axes; missing populations or zero Ninja tails mask ratios. All five configured reducers remain visible. No performance acceptance.

Configured reducer	Result state	Available / unavailable	Zero / infinite	Tail bounds at 1e-3
CutTools	complete	100 / 0	0 / 0	0%–0%
Ninja	complete	100 / 0	0 / 0	0%–0%
COLLIER	complete	100 / 0	2 / 0	0%–0%
NinjaRS	complete	100 / 0	0 / 0	0%–0%
TIRust	complete	100 / 0	1 / 0	0%–0%

Indexed PDF export: `stability_appendix_c/raw-dp-pilot-5-figures-1/figures/ew_n2_epem_mupmum.pdf`

25 Raw stability pilot · $u\bar{u} \to ZZgg$

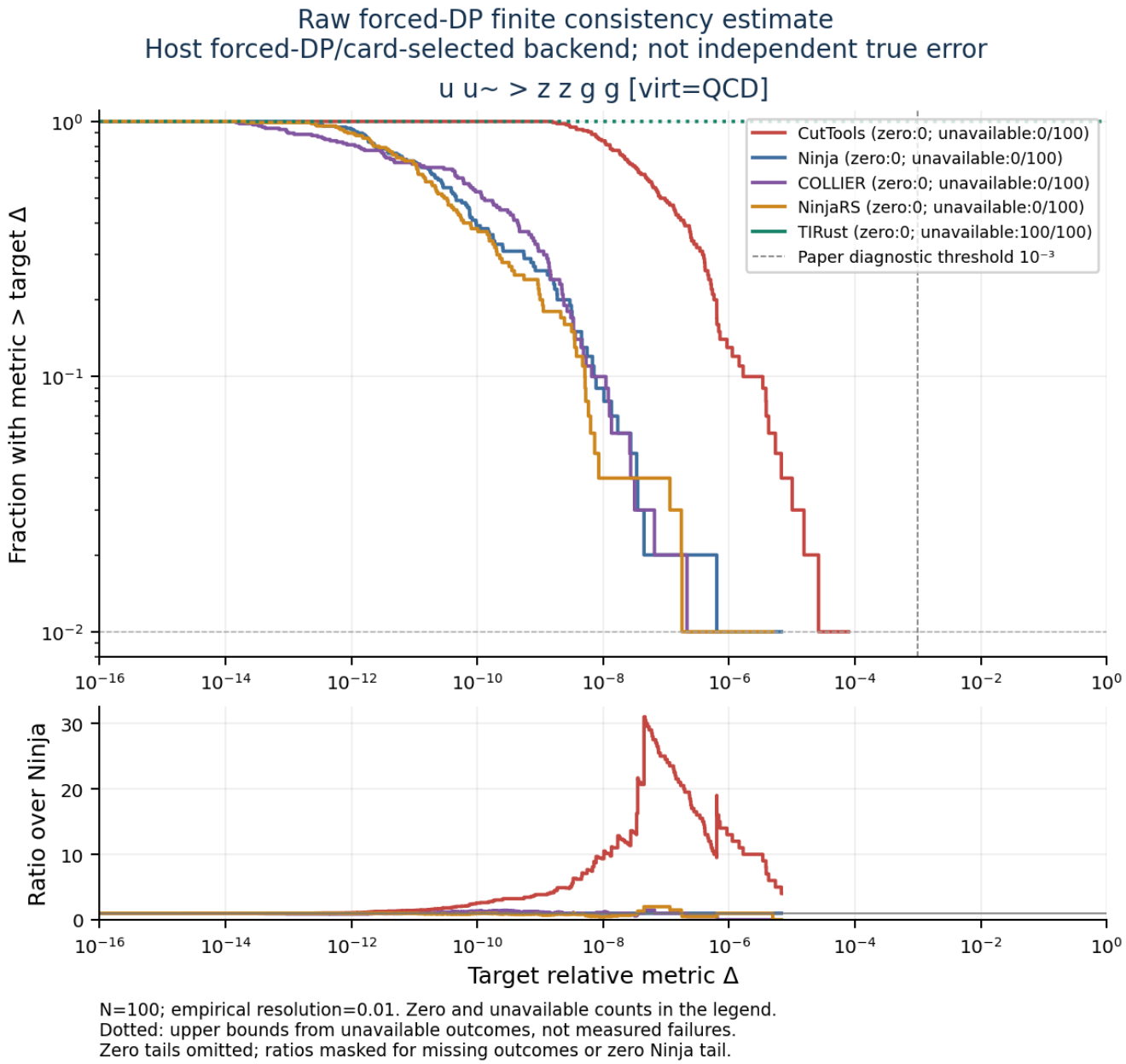


Figure 12: Host forced-DP/card-selected backend; actual backend and internal precision unobserved. Three-probe consistency on 100 cut-qualified points (1% empirical resolution), not true error. Dotted curves are upper bounds for unavailable outcomes, not measured failure fractions. Zero tails are omitted on log axes; missing populations or zero Ninja tails mask ratios. All five configured reducers remain visible. No performance acceptance.

Configured reducer	Result state	Available / unavailable	Zero / infinite	Tail bounds at 1e-3
CutTools	complete	100 / 0	0 / 0	0%–0%
Ninja	complete	100 / 0	0 / 0	0%–0%
COLLIER	complete	100 / 0	0 / 0	0%–0%
NinjaRS	complete	100 / 0	0 / 0	0%–0%
TIRust	failed	0 / 100	0 / 0	0%–100%

TIRust's stream failed/incomplete: all 100 point diagnostics remain unavailable, so the tail is unknown in [0, 1], not a measured 100% failure fraction. There are 254 unchecked primary status-110 returns and 46 unexecuted primary calls, with 212 active-group failure calls and 254 latched calls. The 1190.177 s deadline/cleanup duration is a diagnostic observation, not precise benchmark timing.

Indexed PDF export: `stability_appendix_c/raw-dp-pilot-5-figures-1/figures/qcd_n4_uux_zzgg.pdf`

26 Raw pilot failure and call accounting

Primary return-code histograms use order-zero calls from the three probes. Primary outcome counts retain all 300 intended calls per lane; unavailable or unexecuted outcomes are not silently dropped. Bootstrap return codes are separate. Group and floating-point-latch counts are observed primary-call diagnostics; they are not measured point-failure fractions. All error records, raw diagnostics and unavailable-point reasons remain bound in the dossier. Sticky floating-point-latch observations may persist into later calls; their count is not the number of independent exceptions.

26.1 $gg \to t\bar{t}$

Reducer	Primary RET_CODE	Bootstrap RET_CODE	Primary outcomes / 300	Group failure counts
CutTools	{"110": 300}	{"110": 1, "130": 4}	{"diagnostic_available": 300}	0 / 0 / 0 / 0 / 0
Ninja	{"110": 300}	{"110": 1, "130": 4}	{"diagnostic_available": 300}	0 / 0 / 0 / 0 / 0
COLLIER	{"110": 300}	{"110": 1, "130": 4}	{"diagnostic_available": 300}	0 / 0 / 0 / 0 / 0
NinjaRS	{"110": 300}	{"110": 1, "130": 4}	{"diagnostic_available": 300}	0 / 0 / 0 / 0 / 0
TIRust	{"110": 300}	{"110": 1, "130": 4}	{"diagnostic_available": 300}	0 / 0 / 0 / 0 / 0

26.2 $u\bar{u} \to Zgg$

Reducer	Primary RET_CODE	Bootstrap RET_CODE	Primary outcomes / 300	Group failure counts
CutTools	{"110": 300}	{"110": 1, "130": 4}	{"diagnostic_available": 300}	0 / 0 / 0 / 0 / 0
Ninja	{"110": 300}	{"110": 1, "130": 4}	{"diagnostic_available": 300}	0 / 0 / 0 / 0 / 0
COLLIER	{"110": 300}	{"110": 1, "130": 4}	{"diagnostic_available": 300}	0 / 0 / 0 / 0 / 0
NinjaRS	{"110": 300}	{"110": 1, "130": 4}	{"diagnostic_available": 300}	0 / 0 / 0 / 0 / 0
TIRust	{"110": 300}	{"110": 1, "130": 4}	{"diagnostic_available": 72, "rejected": 228}	0 / 15 / 0 / 0 / 228

26.3 $e^+e^- \to \mu^+\mu^-$

Reducer	Primary RET_CODE	Bootstrap RET_CODE	Primary outcomes / 300	Group failure counts
CutTools	{"110": 300}	{"110": 1, "130": 4}	{"diagnostic_available": 300}	0 / 0 / 0 / 0 / 0
Ninja	{"110": 300}	{"110": 1, "130": 4}	{"diagnostic_available": 300}	0 / 0 / 0 / 0 / 0
COLLIER	{"110": 300}	{"110": 1, "130": 4}	{"diagnostic_available": 300}	0 / 0 / 0 / 0 / 0
NinjaRS	{"110": 300}	{"110": 1, "130": 4}	{"diagnostic_available": 300}	0 / 0 / 0 / 0 / 0
TIRust	{"110": 300}	{"110": 1, "130": 4}	{"diagnostic_available": 300}	0 / 0 / 0 / 0 / 0

26.4 $u\bar{u} \to ZZgg$

Reducer	Primary RET_CODE	Bootstrap RET_CODE	Primary outcomes / 300	Group failure counts
CutTools	{"110": 300}	{"110": 1, "130": 4}	{"diagnostic_available": 300}	0 / 0 / 0 / 0 / 0
Ninja	{"110": 300}	{"110": 1, "130": 4}	{"diagnostic_available": 300}	0 / 0 / 0 / 0 / 0
COLLIER	{"110": 300}	{"110": 1, "130": 4}	{"diagnostic_available": 300}	0 / 0 / 0 / 0 / 0
NinjaRS	{"110": 300}	{"110": 1, "130": 4}	{"diagnostic_available": 300}	0 / 0 / 0 / 0 / 0
TIRust	{"110": 254}	{"110": 1, "130": 4}	{"rejected": 254, "unexecuted": 46}	0 / 212 / 0 / 0 / 254

Group columns are missing coverage / active-group failure / nonfinite coefficients / nonfinite results / floating-point failure latch. These categories may overlap. This diagnostic does not promote a scientific or performance gate.

27 Current-source progress · 0fcd2cd873d6

Campaign entry-latch-cache-guard-1

The preceding report and plots retain campaign invariant-symmetry-1, source 2c13246ffa63, and published PDF commit 913a746201eb. This supplement records the separately registered source 0fcd2cd873d6. Its evidence does not change the historical measurements or their source identities.

The generated Fortran now resets the DP/QP software reduction-failure latches at each independent amplitude entry and checks context generation before clearing TIRust events. The source-specific records below qualify their stated numerical scope.

Current source binding	Retained identity
Exporter inventory	afb6e0bea64ea0932751ff4869c206bcea9477d40a39a54bb99d0f06d85b75a6
Numerical identity	3c13ad88b825d9d19b1030b62b317c0f31759e002d8dfdb9050142301a7cae24
Native DSO	eb3d67fdc99b742b1c5d66101522c8b8fb97e74b0b71ac69fd4452fb60c42af9

27.1 Frozen ordinary numerical coverage

11 of 11 cases have closed records: 66 of 66 lanes, with 100 saved points per lane. 11 cases pass the 3e-6 regression criterion.

Table 7: Source 0fcd2cd873d6: retained ordinary numerical records. Regression and ordinary-target outcomes are shown separately.

Process	Status	Lanes	3e-6	1e-8	Accepted DP / QP
u u~ > z [virt=QCD] main	qualified	6 / 6	pass	pass	600 / 0
g g > t t~ [virt=QCD] main	qualified	6 / 6	pass	pass	600 / 0
u u~ > z g g [virt=QCD] main	qualified	6 / 6	pass	miss	587 / 13
u u~ > z z g g [virt=QCD] main	qualified	6 / 6	pass	miss	483 / 117
u d~ > w+ [virt=QED] main	qualified	6 / 6	pass	pass	600 / 0
e+ e- > mu+ mu- [virt=QED] main	qualified	6 / 6	pass	pass	600 / 0
e+ e- > t t~ a [virt=QED] main	qualified	6 / 6	pass	pass	600 / 0
e+ e- > mu+ mu- a a [virt=QED] main	qualified	6 / 6	pass	miss	526 / 74
u u~ > g a [virt=QCD QED] control	qualified	6 / 6	pass	pass	600 / 0
g g > h [noborn=QCD] control	qualified	6 / 6	pass	pass	600 / 0
g g > h h [noborn=QCD] control	qualified	6 / 6	pass	pass	600 / 0

Pending denotes absence of selected closed evidence; it makes no claim about active, later or unselected work. Accepted DP/QP totals cover baseline CutTools and candidate reducers 1/6/7/8/9. They describe final accepted outcomes and do not enumerate internal retries. Software/hardware exception flags were not recorded. These records supply no paired runtime acceptance.

Target-miss case	Tool	Scalar misses	B / F / 1 / 2	Saved points	Max. finite residual
qcd_n3_uux_zgg	6	56	0 / 56 / 0 / 0	28	2.806×10^{-6}
qcd_n3_uux_zgg	7	56	0 / 56 / 0 / 0	28	2.806×10^{-6}
qcd_n3_uux_zgg	8	56	0 / 56 / 0 / 0	28	2.806×10^{-6}
qcd_n3_uux_zgg	9	56	0 / 56 / 0 / 0	28	2.806×10^{-6}
qcd_n4_uux_zzgg	6	88	0 / 88 / 0 / 0	44	4.129×10^{-7}

Target-miss case	Tool	Scalar misses	B / F / 1 / 2	Saved points	Max. finite residual
qcd_n4_uux_zzgg	7	80	0 / 80 / 0 / 0	40	4.151×10^{-7}
qcd_n4_uux_zzgg	8	80	0 / 80 / 0 / 0	40	4.129×10^{-7}
qcd_n4_uux_zzgg	9	78	0 / 78 / 0 / 0	39	4.129×10^{-7}
ew_n4_epem_mupmumaa	6	12	0 / 10 / 2 / 0	5	1.144×10^{-7}
ew_n4_epem_mupmumaa	7	10	0 / 8 / 0 / 2	5	1.144×10^{-7}
ew_n4_epem_mupmumaa	8	14	0 / 10 / 2 / 2	5	1.144×10^{-7}
ew_n4_epem_mupmumaa	9	8	0 / 8 / 0 / 0	4	1.144×10^{-7}

B / F / 1 / 2 gives the Born, finite, single-pole and double-pole miss counts. Their sum equals the scalar count. The ordinary-target counts retain both order rows at each saved point; they are not counts of distinct physical failures. Residuals compare against the source-matched baseline CutTools result and supply no independent truth estimate. Born and both pole columns pass 1e-8 in the retained QCD n3 miss case; its misses are finite only.

The EW n3 snapshot binds 375 unchanged inherited baseline objects, 426 generated source pins and the donor compiler flags. The original baseline full-build receipt was not located in this audit; this does not demonstrate a fresh baseline full rebuild. Candidate exports and numerical objects are fresh. Numerical agreement and this baseline provenance limitation remain separate.

The QCD n4 snapshot binds 155 unchanged inherited baseline objects, 206 generated source pins and the donor compiler flags. The original baseline full-build receipt was not located in this audit; this does not demonstrate a fresh baseline full rebuild. Candidate exports and numerical objects are fresh. Numerical agreement and this baseline provenance limitation remain separate.

The EW n4 snapshot binds 320 unchanged inherited baseline objects, 371 generated source pins and the donor compiler flags. The original baseline full-build receipt was not located in this audit; this does not demonstrate a fresh baseline full rebuild. Candidate exports and numerical objects are fresh. Numerical agreement and this baseline provenance limitation remain separate.

The EW n4 original attempt was interrupted for a user-requested daemon update after TIRust saved point 74. Its five completed baseline/CutTools/Ninja/COLLIER/NinjaRS lanes remain selected. The interrupted TIRust stream is retained but excluded from qualification; a separate full 100-point TIRust attempt supplies the sixth selected lane. No partial streams were concatenated, and the original interrupted stage has not been relabelled as passed.

28 Cache-guard scope · 0fcd2cd873d6

Scope: guard alone. The following paired observations compare the guarded candidate with the original guard-off code. This comparison excludes the entry-latch reset. Wrapper rows measure lifecycle operations alone; full-amplitude rows cover one shared-host QCD n1 process. These measurements do not measure the combined source changes or establish general performance, campaign acceptance or precise local performance.

Table 8: All retained guard-only scopes, with 7 pairs per scope. Ratios above 1 indicate overhead in that scope. The 95% paired intervals describe uncertainty.

Guard-only scope	Guarded / original	95% interval
Empty wrapper; no context	0.663651	0.662873 – 0.664430
Active wrapper; context registered	1.007618	1.002979 – 1.012277
Full QCD n1; CutTools	0.996826	0.994922 – 0.998734
Full QCD n1; TIRust	0.996882	0.990792 – 1.003011

Retained numerical-byte parity: **pass**. Active-wrapper overhead is approximately 0.76%. Cache-operation timing and byte parity do not establish independent physical accuracy, complete campaign coverage or precise local performance acceptance.

28.1 Fresh local qualification · 0fcd2cd873d6

All 66 fresh local numerical lanes remain **pending**. The closed records above and the guard-only measurements are separate from this local prerequisite.

Follow Hua-Sheng's current-source local timing handoff for a fresh native installation and receipt, local preparation, full saved-point numerical replay, paired timing and a separate local report.

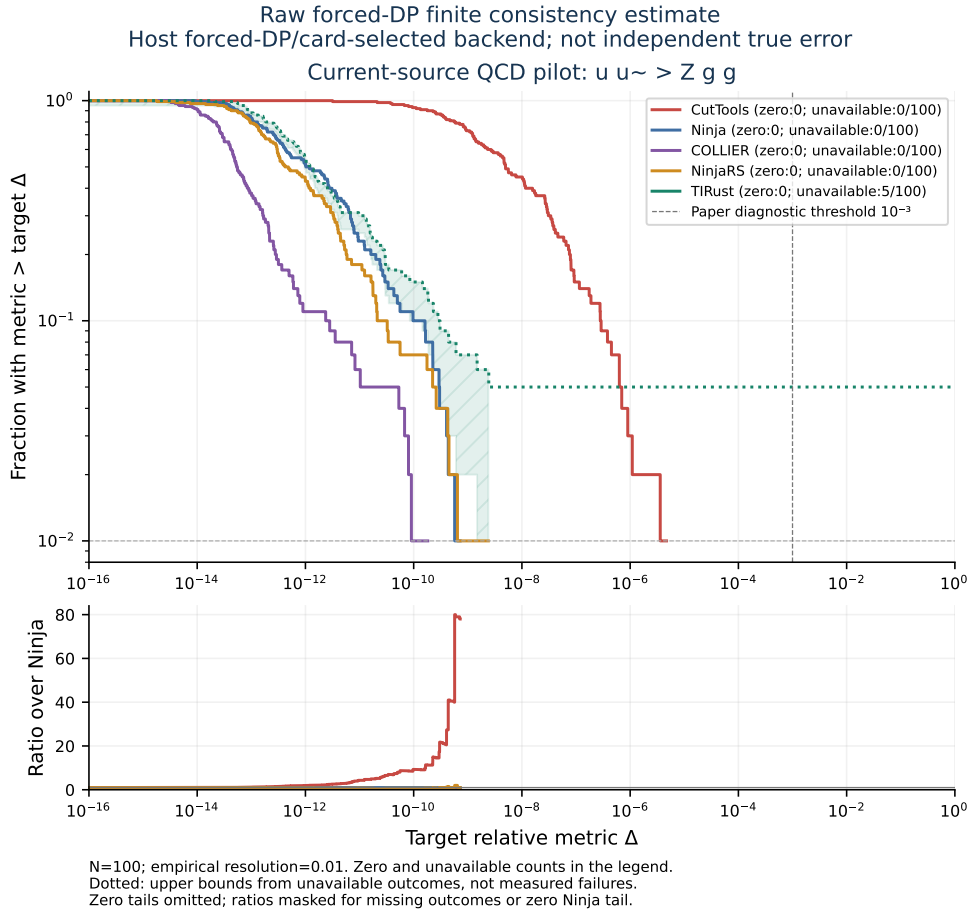
The full-stream precision TODO retains the remaining precision and reporting work.

Overall delivery: **incomplete**. Precise local performance acceptance remains deferred to Hua-Sheng.

29 Raw stability pilot · 0fcd2cd873d6

Current-source QCD n3 · 100 cut-qualified physical points · card-selected reducers 1/6/7/8/9

Tool	Execution	Available / 100	Unavailable	Rejected probes	Tail at 1e-3
1	complete	100	0	0	0% – 0%
6	complete	100	0	0	0% – 0%
7	complete	100	0	0	0% – 0%
8	complete	100	0	0	0% – 0%
9	complete	95	5	15	0% – 5%



Current-source QCD pilot: u u~ > Z g g. Host forced-DP with card-selected backends; return code 110 does not identify the actual reducer or its internal precision. The 100 cut-qualified physical points give 1% empirical resolution. Dotted curves bound unavailable outcomes; they are not measured failure fractions. Incomplete populations and zero Ninja exceedances mask ratios. Internal consistency is not independent true error; no performance claim is made.

The same three-probe finite/Born relative-spread diagnostic covers 1500 primary and 25 separate bootstrap calls. TIRust's five unavailable points (25, 34, 45, 70, 84) retain 15 rejected probes and 27 failed active-group observations. Five subsequent healthy calls clear incoming software latches under the source-qualified entry reset; this does not qualify the failed calls. Exact legacy annotations remain retained. Hardware IEEE state was not observed. The 1e-3 TIRust fraction is unknown within 0–5%, and its Ninja ratio is masked.

Raw data and all outcomes · Figure and statistical bounds. This selected study adds no ordinary-case, true-error or performance acceptance.

CMS stress diagnostic: closed outcomes and limits

Private prototype; no production promotion. The source-derived Fortran invariant certificate has conditional QP pole support at the forward CMS fixture. Its complete stress comparison remains failed and incomplete; the finite term remains unresolved. These observations do not qualify normal MadLoop stability/rescue or precise performance.

The fixture is $e^+e^- \to \mu^+\mu^-$ with QED virtual corrections and the existing complex-mass Gmu model. The prototype changes isolated generated Fortran; the original TIRust shared library is unchanged. Fixed mode 4 returns 120, an unchecked return. The eligible workers have both post-call software failure flags clear; this supplies no independent host accuracy guarantee.

Table 1: Execution completeness, eligibility and numerical agreement are separate outcomes.

Closed stage	Outcome	Scope retained
First build attempt	Failed	Controller `KeyError('size')`; zero native commands, loader queries or amplitudes.
Successor ordinary	8 / 8 eligible	Point 1, original/candidate × CutTools/TIRust × DP/QP. Ordinary target met; private fixed-mode evidence only.
Full stress	Incomplete; exit 1	Eight point-20 workers complete. Ninth worker, original CutTools DP at point 21, times out after 300 s, is terminated by signal 15 and reaped. Seven workers never launch.
Point-20 subset	5 / 8 eligible	Three failed TIRust workers' buffers excluded. Candidate TIRust QP is eligible but its comparison with CutTools QP fails; no complete stress result exists.
Legacy reference probe	0 / 2 eligible	NinjaRS and Ninja complete their protocols; both fail their scientific eligibility gates. No coefficient comparison, including bootstrap, is interpreted.

Original/candidate CutTools controls agree exactly. Original TIRust DP/QP and candidate TIRust DP retain software failures and W03 warnings at point 20. Candidate TIRust QP has clear post-call flags but differs from CutTools, with maximum recorded `abs(a-b)/max(abs(a),abs(b))` of 1.237729 across the compared coefficients. The later universal-pole audits below support its poles under their reconstruction assumptions; they do not turn that failed full-coefficient comparison or the incomplete stage into a pass.

Table 2: Both probes use the legacy kinematic helper; matched native input geometry with the prototype is unestablished.

Legacy QP probe	Eligible	Why no coefficient arithmetic is allowed
NinjaRS (8)	No	Five point-20 box groups return status 1: 3, 4, 8, 12, 24. The software QP failure flag is set.
Ninja (6)	No	Adapter statuses are zero but final finite and both pole words are nonfinite. The host software DP failure flag is set despite zero observed DP adapter calls.

Their 104 completed adapter calls establish the QP interface route (storage width 128, precision digits 113), not every reducer's internal working precision. No scalar cutoff was changed and no particular scalar-call cause is proven.

Independent universal-pole comparisons

For four massless unit-charge external fermions, the double-pole prediction is $A_{-2} = -4B\frac{\alpha}{2\pi}$. The source-derived single-IR-pole prediction is $A_{-1} = B\frac{\alpha}{2\pi}\left[-6 - 2\sum_{i<j} Q_i Q_j \ln\left(\frac{|2k_i\cdot k_j|}{\mu^2}\right)\right]$, with all-outgoing charges $(-1,+1,+1,-1)$ and $\mu^2 = 10^6$. The retained FKS convention cancels the auxiliary QES scale algebraically; these checks introduce no scale-varied amplitude evaluations.

Table 3: Universal QED double pole.

Point	Eligible fixed-QP worker	Predicted pole	Observed pole	Relative residual
1†	CutTools QP control	-7.02262×10^{-5}	-7.02262×10^{-5}	3.674×10^{-16}
1†	Private candidate TIRust QP	-7.02262×10^{-5}	-7.02262×10^{-5}	1.857×10^{-17}
20	CutTools QP control	-1.62777×10^{-4}	-8.13900×10^{-5}	5.000×10^{-1}
20	Private candidate TIRust QP	-1.62777×10^{-4}	-1.62777×10^{-4}	2.679×10^{-16}

Table 4: Universal QED single IR pole of the intended renormalized amplitude.

Point	Eligible fixed-QP worker	Predicted pole	Observed pole	Relative residual
1†	CutTools QP control	-1.44006×10^{-4}	-1.44006×10^{-4}	5.481×10^{-17}
1†	Private candidate TIRust QP	-1.44006×10^{-4}	-1.44006×10^{-4}	2.430×10^{-16}
20	CutTools QP control	-4.96751×10^{-3}	-2.33547×10^{-3}	5.299×10^{-1}
20	Private candidate TIRust QP	-4.96751×10^{-3}	-4.96751×10^{-3}	5.231×10^{-16}

† Point 1 rows are the selected point-20 workers' bootstrap, not additional independent physical samples. Two exactly duplicated answer orders are collapsed once. Current candidate CutTools controls match the displayed original CutTools rows exactly.

At point 20, private candidate TIRust QP agrees with the reconstructed double and single IR poles to about 2.68×10^{-16} and 5.23×10^{-16} respectively. Original CutTools QP differs by about 50.0% and 53.0%. This is conditional coefficient consistency; the finite coefficient and full-amplitude accuracy remain unresolved.

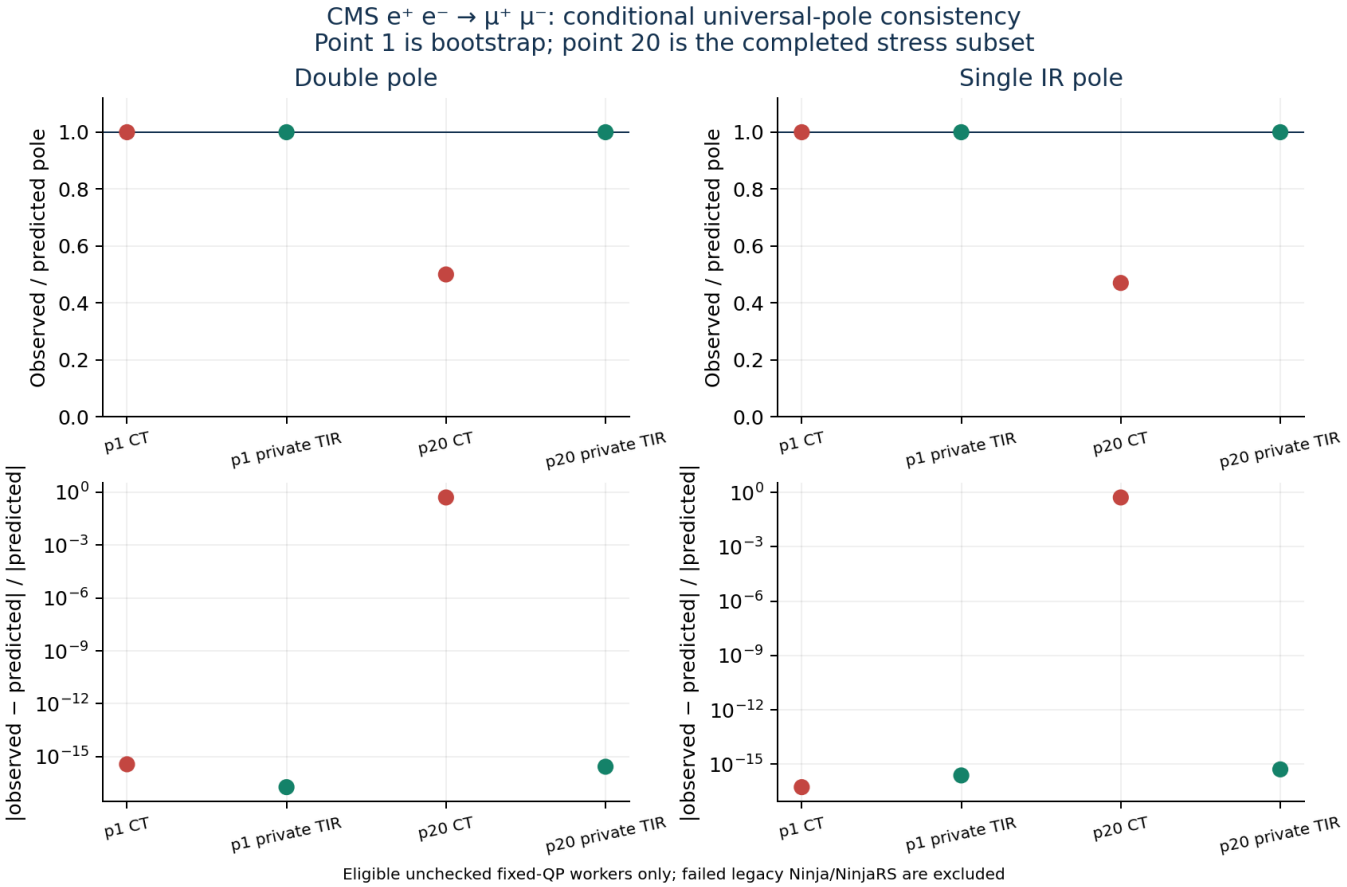


Figure 1: Observed/predicted poles and relative residuals for the eight eligible observations. Failed legacy probes are excluded, not assigned zero.

Assumptions that limit the conclusion

The coupling is reconstructed as $\alpha = \sqrt{2}G_F \frac{\left|\mu_W^2\left(1-\frac{\mu_W^2}{\mu_Z^2}\right)\right|}{\pi}$, $\mu_V^2 = M_V^2 - iM_V\Gamma_V$, with the existing `GFRedefinitionChoice=1` and the retained model's decimal pi literal. Parsing card values through binary64 is an explicit assumption: retained packets do not contain the card-reader/MP-widening implementation or observed runtime coupling bits. The original alpha and pole calculation uses 70-digit Decimal working arithmetic; independent pole-factor and scale-cancellation reviews use 90 digits. Neither measures native coupling precision or emulates every Fortran rounding step.

The single-pole calculation uses earlier closed group-3 QP trace leaves for points 1/20 and exact rational dot products of those words. Their original point-20 stress run remains failed. Matching cards, point streams and an unchanged momentum-source path justify the documented reuse; the current prototype made no new external-word capture. Restored QP leaf components must not be equated with the un-restored binary64 input stream. The prediction applies to the intended UV-renormalized full `ANS`, with counterterms and UV+IR selection. Inclusion of counterterms is source-linked; UV cancellation and every model counterterm were not separately tested. This is not a bare tensor `IR1` test.

Reproduction and remaining acceptance

The pure generator authenticates the four closed public indexes and every indexed payload, joins the selected eligible pole words to the closed parent, and preserves failed reference buffers without decoding or arithmetic. It does not re-execute the packets' broader dependency verifiers, rehash today's external binaries, or reparse all historical generated sources.

Public packet seals: double pole `1565ac55fe6a…`; single pole `99603d41d662…`; private amplitude `c04a73c86f6a…`; legacy references `248bd61e987a…`. Complete seals and selected leaf hashes are in `data.json`.

Local precise timing TODO — Hua-Sheng. Use identical local builds, points, models and cards; run at least seven alternating paired batches of about one second. Require the upper 95% runtime ratio bound to be ≤1.05 for every main process, with existing-backend regressions and native-DP/rescued cohorts reported separately. Current stage elapsed times are accountability records, not measurements satisfying that requirement.

Later matched-input QP comparison

Later evidence; the preceding three pages retain their earlier chronology. Two eligible private fixed-QP workers now agree at the same CMS point 20 in Born, finite, single- and double-pole outputs. The maximum retained `abs(a-b)/max(abs(a),abs(b))` is 2.014×10^{-16}. This is matched-input agreement, not an analytic finite oracle or an algorithmic winner. The private input-policy change is not promoted.

The closed comparison uses backend 8 (NinjaRS) and backend 9 (TIRust), one point-1 bootstrap and one point-20 measurement each: four outer calls and 104 native adapter calls. All native statuses are zero, both whole-worker eligibility gates and bootstrap controls pass, and post-call software DP/QP failure flags are clear. Fixed return 120 remains an unchecked evaluation.

What was matched and what changed

For all 52 paired native requests, the ordered shifts, full scalar-invariant matrix, complex squared masses, scale and canonical numerator words match exactly. Explicit routing/producer metadata and entry/use environment tokens also match. The ordered group-3 entries $(2, 4)$ and $(4, 2)$ actually use the certified direct spacelike invariant. QP interface storage is 128 bits with 113 precision digits; this does not assert every reducer's internal precision.

Backend 8 here uses a separate private input-policy derivative. It shares the candidate's unchanged certified invariant helper, retains a provably negative raw invariant where a qualified composite pair would otherwise be overwritten by the legacy zero branch, aligns the authoritative-upper symmetry policy, and evaluates the scale as `REAL(MU_R,KIND=KI_QNRS)**2`. Original legacy 8 retains its original helper and `REAL(MU_R**2,KIND=KI_QNRS)` route. No library, scalar cutoff, physical card or production backend is changed by this test. The earlier failed original-8/Ninja-6 reference probe remains failed.

Table 1: Point-20 retained binary64 outer-output words from fixed-QP evaluations. Both answer orders agree; exact duplicate orders are counted once.

Coefficient	Private NinjaRS 8	Private TIRust 9	ULP gap	Relative residual
Born	3.38643×10^{-2}	3.38643×10^{-2}	0	0
Finite	-3.44596×10^{-2}	-3.44596×10^{-2}	1	2.014×10^{-16}
Single pole	-4.96751×10^{-3}	-4.96751×10^{-3}	1	1.746×10^{-16}
Double pole	-1.62777×10^{-4}	-1.62777×10^{-4}	0	0

Born and double-pole words are bit-identical. Finite and single-pole outputs differ by one ULP of the retained binary64 word. All coefficients and both orders satisfy the existing regression criterion and the 10^{-8} target. Native success alone was not used to admit coefficient arithmetic.

Connection to the earlier pole audits and remaining scope

The current backend-9 output words match the prior eligible private candidate exactly: sixteen Born/Laurent words across bootstrap/measurement and two orders, plus four ancillary diagnostic words. Source/runtime-provider bindings are retained. Historical native input words were not captured, so this output parity does not retroactively measure equality of the earlier native inputs. The preceding conditional pole audits concern those same observed outputs; their coupling-reconstruction, restored-leaf and UV-cancellation limitations remain in force. No failed reference coefficient is selected or decoded.

Still unqualified. Two-reducer agreement does not establish independent full-amplitude truth, a general source/exporter fix, normal checked MadLoop stability/rescue, generic process/topology coverage or performance. The earlier incomplete stress stage is not reclassified. Precise local timing remains Hua-Sheng's TODO with the agreed per-process upper-95% ratio bound ≤1.05.

Closed run `52faa470d8be…`; comparison `af4b58a379ac…`; root review `9ca6601ab14a…`; prior/current word parity `5f9f4cc6c8d2…`. Full pins and the public evidence join are retained in the companion data/checker. Closed public evidence.

Five-backend Standard Model timings

Descriptive shared-host measurements. These full-amplitude timings compare explicit CutTools, Ninja, COLLIER, NinjaRS and TIRust on the selected saved cohorts. There is no CPU reservation and no confidence-interval or final local-performance acceptance claim. The earlier 39 pages retain their separately bound scientific and timing history.

GNU Fortran (GCC) 15.2.0; AMD EPYC 9754 128-Core Processor; affinity CPU 12; one-thread environment on a shared, unreserved host.

Elapsed time is the unchanged Fortran `SYSTEM_CLOCK` wall clock, divided by the actual `points × loops`. Each backend has ten position-balanced cycles (five Latin positions followed by reversal), with loops fixed after two retained calibrations. The nominal batch target is two seconds; actual batch durations and counts remain in the closed evidence. The table reports the median and **unscaled** median absolute deviation, not an error bar on the mean. Ratios are medians of ten within-cycle backend / NinjaRS8 ratios. Actual timed batches range from 1.95 to 3.14 seconds; integer full-cohort traversal counts can exceed the nominal target.

Table 1: Warmed full-amplitude wall time. Ratios below one mean that backend is faster than NinjaRS8. All actual saved-cohort sizes and accepted output-call counts are retained.

Process · cohort	Reducer	READY s	Wall ms/point · median ± MAD	Ratio / 8	Accepted DP / QP	Status
$u\bar{u} \to Z$ · QCD N=100	CutTools	0.0343	$0.127 \pm 9.53 \times 10^{-4}$	×1.48	157000 / 0	3e-6 pass
	Ninja	0.0282	$0.0876 \pm 9.14 \times 10^{-4}$	×1.01	237000 / 0	3e-6 pass
	COLLIER	0.0358	$0.116 \pm 7.30 \times 10^{-4}$	×1.37	173000 / 0	3e-6 pass
	NinjaRS	0.0297	$0.0854 \pm 1.78 \times 10^{-3}$	×1.00	239000 / 0	3e-6 pass
	TIRust	0.0166	$0.0946 \pm 1.36 \times 10^{-3}$	×1.11	214000 / 0	3e-6 pass
$gg \to t\bar{t}$ · QCD N=100	CutTools	0.615	5.55 ± 0.0559	×1.30	4000 / 0	3e-6 pass
	Ninja	0.447	4.23 ± 0.0255	×0.998	5000 / 0	3e-6 pass
	COLLIER	0.583	5.40 ± 0.0319	×1.27	4000 / 0	3e-6 pass
	NinjaRS	0.453	4.25 ± 0.0325	×1.00	5000 / 0	3e-6 pass
	TIRust	0.482	4.46 ± 0.0249	×1.05	5000 / 0	3e-6 pass
$e^+e^- \to \mu^+\mu^-$ · EW N=100	CutTools	0.442	4.12 ± 0.0144	×1.21	5000 / 0	3e-6 pass
	Ninja	0.365	3.31 ± 0.0357	×0.975	6000 / 0	3e-6 pass
	COLLIER	0.488	4.49 ± 0.0374	×1.33	5000 / 0	3e-6 pass
	NinjaRS	0.376	3.38 ± 0.0155	×1.00	6000 / 0	3e-6 pass
	TIRust	0.366	3.54 ± 0.0236	×1.04	6000 / 0	3e-6 pass
$u\bar{u} \to e^+e^-\mu^+\mu^-$ · QCD N=100	CutTools	1.53	14.9 ± 0.0813	×1.03	2000 / 0	3e-6 pass
	Ninja	1.47	14.4 ± 0.127	×1.000	2000 / 0	3e-6 pass
	COLLIER	1.55	15.0 ± 0.174	×1.05	2000 / 0	3e-6 pass
	NinjaRS	1.48	14.4 ± 0.0872	×1.00	2000 / 0	3e-6 pass
	TIRust	1.49	14.5 ± 0.0970	×1.01	2000 / 0	3e-6 pass

The separate ordinary 1e-8 target passes in 20 / 20 lanes.

READY time includes the first amplitude and initialization/learning, excluding SETPARA and process startup; it is separate from warmed timing. Accepted DP/QP counts cover timed output calls and do not recover unobserved internal retries or floating-point exception flags. Numerical checks and timing status are separate; failed or unavailable comparisons remain explicit. Coefficient checks cover the retained final 100-point traversal of learning, each calibration and each timed batch; accepted-status counts cover all timed repetitions.

The $u\overline{u} \to Z$ cohort varies physical on-shell Z boosts. The other three selected cohorts use 1 TeV. These labels follow actual saved-point physics and generation records, not the global configuration energy field.

Timing comparison and scope

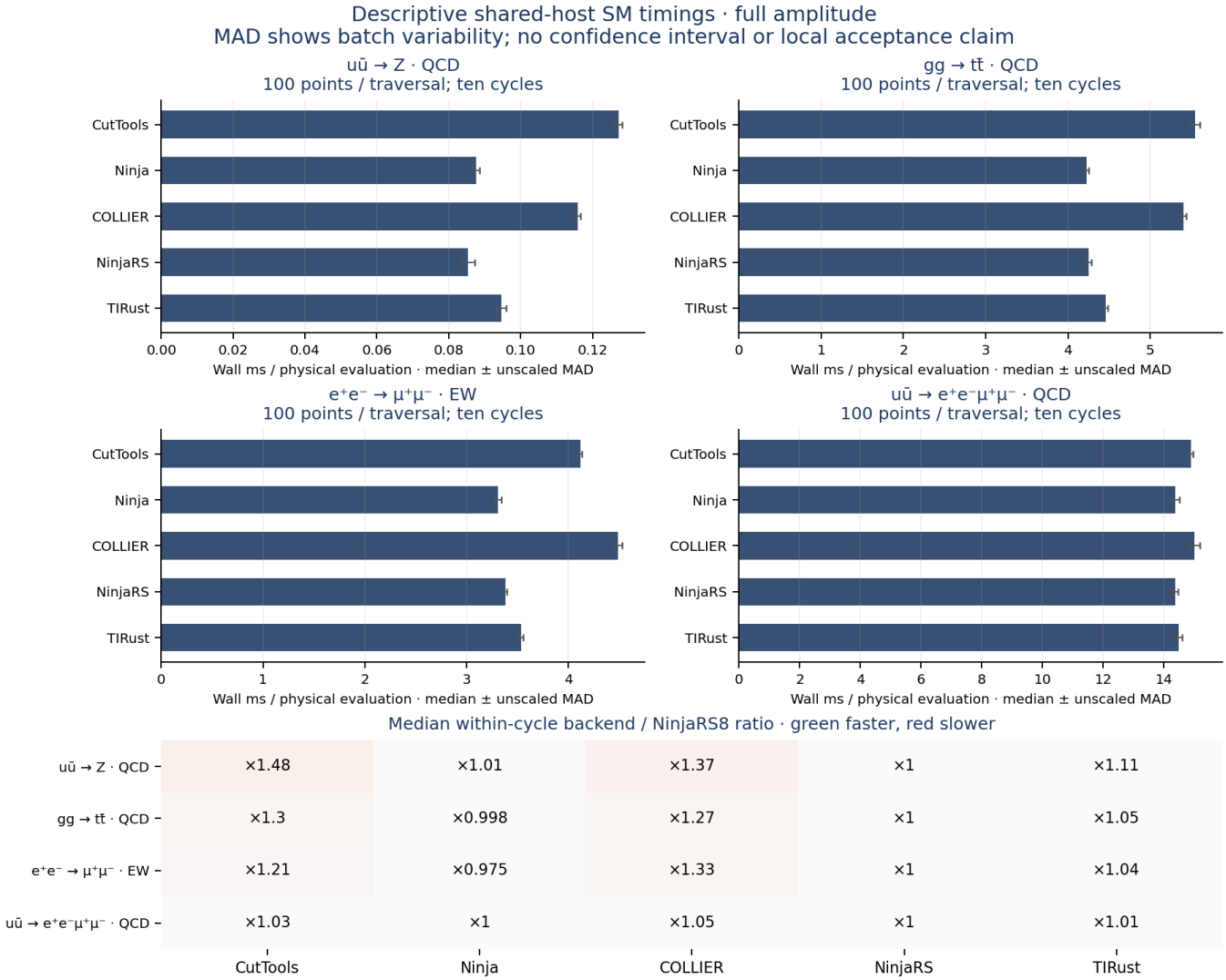


Figure 1: Absolute warmed wall timing for each case and within-cycle ratios relative to NinjaRS8. Whiskers show unscaled MAD, not confidence intervals. Green means a smaller backend/NinjaRS8 ratio, red a larger ratio. Unavailable cells are not zero.

The fresh four-lepton QCD $2 \to 4$ generation has maximum loop N=4 and rank=3, across 12 loop-call groups, as reparsed from its compiled source text. Four final particles do not imply six propagators in each loop. The separately validated ZZgg process retains its full numerical coverage and precise local timing TODO. These shared-host comparisons do not replace the full registered numerical matrix or Hua-Sheng's precise local per-process timing campaign. The agreed upper-95% runtime-ratio bound ≤1.05, source-matched baseline/rescue-chain comparison and existing-backend regressions remain separate acceptance items.

Separate private checked-CMS20 result. One later private CMS20 fixture with cache disabled passes automatic stability checks and genuine QP rescue (return323): six failed DP group reductions remain retained; all52 QP reductions succeed. The earlier failed controller attempt remains failed. This is not generic or production qualification, enabled-cache qualification, independent full-amplitude truth, or a performance result. Closed evidence. This does not qualify the production timing inputs or generic execution.

Closed timing evidence and reproduction commands: Closed timing capsules; see the accompanying README and scientific-selection.json. Clock, source/runtime/card/cohort pins, actual durations, calibration and all accepted-status/precision histograms are retained with that packet.

Additional implementation qualifications

Unpromoted experiments; delivery remains incomplete. These results qualify individual experimental changes. They do not change the production source, registered campaign, or acceptance status above.

The optimized QP caller currently squares denominator masses in binary64 before widening. An experimental change across five source paths passes the model's QP masses and squares them at QP only when backend 9 is actually selected. In the QCD process $u\bar{u} \to Zgg$, original and candidate each complete normal learning and 100 saved final points with 87 DP and 13 naturally rescued QP answers. All 800 amplitude-coefficient words and 200 accuracy words match exactly with native integral caching both disabled and enabled. Native-input observation confirms 1,118 corrected 105-bit bottom-mass squares and 72 changed mass-dependent invariants; numerator, momentum-shift and scale records remain identical. Separate recording-stub controls preserve genuine 113-bit Fortran input words. The physical real-model test does not itself supply 113-bit model masses.

A subsequent fresh MG CLI export, complete generated-source review, compilation and 100-point amplitude run reproduce the corrected candidate's coefficient, accuracy, status and observed exception states exactly. This run uses the QP integrand, native cache disabled, and normal independent stability checks. The review checks all 195 generated source/build-input bindings, 25 unchanged public routines, nine private QP routines and 116 guarded calls; every changed scientific source is recompiled. The earlier failed CLI startup and the historical generation result marked `generated_pending_source_review` remain retained. The later chain supplies source-to-caller correspondence without rewriting those outcomes. Electroweak, loop-induced, complex-mass, five-backend and independent finite-answer qualification remain open, and the patch is unapplied.

A separate Fortran profile measures the existing phase-space repair in $u\bar{u} \to Z$ with independent stability checks. Across seven four-lane cycles and 358,400 timed full-amplitude calls, the median inclusive DP repair is 6.075 µs within a clock-instrumented 89.914 µs amplitude; its nested MP repair is 5.627 µs. These intervals overlap and must not be added. Instrumented and plain lanes preserve final numerical words and observed exception states. These descriptive shared-host measurements identify a modest part of the total runtime; they demonstrate no optimization gain or formal performance acceptance.

Two standalone Fortran components support further complex-mass research. A bounded integer exact-geometry helper passes 67,920 calls across 20 floating-point-environment cohorts, explicitly refusing unprovable cases. A caller-owned snapshot module passes 1,480 observations, 3,680 API outcomes and 94,720 state-word checks under the same 20 cohorts, with checked and optimized builds agreeing. These tests qualify equality checks and snapshot transport only. Actual constructor ownership, use of the current physical point, final scalar products and the complete complex-mass caller still require numerical qualification.

The preceding five-backend timing tables, including the 2⟶4 case, and Appendix-C-style stability figures retain their original datasets and source identities. Native-DP stability and rescued-answer accuracy remain separate; 100 samples resolve fractions only down to 1%. Hua-Sheng's local paired timing run remains the formal performance TODO. None of these experiments changes a failed or unmeasured main-campaign row into a passing result.

Exact supplementary index hashes and scope are supplied in `qualification-selection.json`. The following links provide raw evidence and reproduction commands.

QP-mass stages · Fresh-CLI followup · Repair profile · Integer equality · Snapshot transport.

Analytic Fortran phase-space repair

One-process shared-host study; patch remains unapplied. Original `0fcd2cd873d6` control versus a same-flag Fortran candidate. Hua-Sheng's precise local performance acceptance remains pending.

The 2⟶1 process $u\bar{u} \to Z$ with QCD corrections uses `loop_sm` and explicitly selected TIRust backend 9. The ordinary sample contains 100 saved longitudinal boosts and retains independent MadLoop stability checks. Its measured tensor topology has three propagators and rank two. Only the Fortran repair unit is replaced; all compiler flags and other numerical objects remain unchanged.

Complete warmed amplitude	Median µs/point	Unscaled MAD µs/point	Accepted 219 calls
Original control	95.3571	0.5529	89,600
Unapplied Fortran candidate	90.1986	1.5712	89,600

Each arm has seven batches of 128 full 100-point traversals (12,800 calls per batch), on nonexclusive CPU 12. MAD is the median absolute deviation of batch time per point; it is unscaled, without conversion to a standard error.

Seven matched-history pairs

Pair	1	2	3	4	5	6	7
Original s	1.1922	1.2200	1.2312	1.2206	1.2276	1.2117	1.2243
Candidate s	1.1578	1.1199	1.1817	1.1487	1.1344	1.1545	1.1831
C/O	0.9711	0.9180	0.9598	0.9411	0.9241	0.9528	0.9663

Order is original⟶candidate for odd pairs and candidate⟶original for even pairs. Every actual batch duration is retained in the machine-readable data.

Median paired C/O: 0.9528. Ratio of the two arm medians: 0.9459, a different statistic. All 14 batches span 1.119886–1.231193 s. All 179,200 timed statuses are accepted DP code 219.

Numerical scope and limits

Two standalone mains complete 544 outer DP/QP repair-wrapper invocations across 136 fixtures per precision and arm. DP outputs match bit for bit; 94 QP fixture results differ in 233 component words while satisfying the frozen physical-residual limits. Refused-input outputs, diagnostics and observed IEEE states preserve the original behavior. These tests do not instrument actual fast-branch coverage.

Fresh learning and final 100-point amplitude runs retain exact coefficient, accuracy and public-status words on all 100 physical points. Each arm returns 200 rows (overall plus the sole split order), containing 800 coefficient and 200 accuracy fields. The 14 timing batches retain only their final 100-point numerical outputs; their status histograms cover every timed call. The physical model's MP mass remains its existing binary64-to-binary128 lift.

This is a descriptive comparison against the current control, separate from the starting-revision default-chain and eight-process acceptance campaign. Broader process/model qualification and an independent finite-answer oracle remain open. Recorded IEEE states establish observational parity, without a trap-freedom claim. The source patch remains unapplied; these results do not promote a backend or establish formal local runtime acceptance.

Reproduction and full-precision values: `README.md`, `derive_timing.py` and `analytic-repair-data.json`. Evidence index `f8bc6ce06201…` timing review `9e10f47d4599…`. Typography and timing palette follow the preserved pyAmpliCol source, revision `7266c1600f6e…`.

Electroweak mass-interface qualification

Ordinary DP regression passes; the threshold failure remains open. The QP-mass source patch is unapplied. These controls establish neither independent physical accuracy nor performance acceptance.

Both studies use `e+ e- > t t~ a [virt=QED]` in the pinned `loop_qcd_qed_sm_Gmu` model, with TIRust backend 9, cache disabled and normal stability handling at `MLStabThres=1e-6`. Fresh CLI generation, generated-source correspondence and compilation are retained for the original integration and the mass prototype. The comparison concerns this one EW process.

Ordinary sample: 100 accepted DP points per arm

Observed per arm	Original	Mass candidate
Physical points	100	100
Accepted DP / QP points	100 / 0	100 / 0
Public accepted status	219	219
Returned coefficient / accuracy slots	800 / 200	800 / 200

All returned coefficient, accuracy, status, history and observed IEEE/software-state words agree exactly. The pair passes both the `3e-6` regression gate and `1e-8` target with zero pair difference. Its maximum reported accuracy estimate is `1.0098069303873654e-7`, below the checked threshold. These are agreement and estimator observations, not an independent finite-answer oracle. No accepted QP point or changed native QP-mass input is observed; unobserved internal QP attempts are not excluded.

Each worker retains 101 initialization calls, including one scientifically unqualified bootstrap. The remaining 100-point warmup and final sample each return 200 overall/sole-order rows; those duplicate rows are not additional physical points. Explicit `LoopPoleMode=0` evaluates full sums and restores stored filter state after each call. These records do **not** qualify persistent filter learning.

Threshold / soft-photon fixture: both arms refuse

Target and observation	Original	Mass candidate
Ordinary control	Accepted 219	Accepted 219
Saved threshold fixture	**Refused 420**	**Refused 420**
Reported threshold accuracy	1	1
QP software latch before → after	0 → 1	0 → 1

The ordinary control and complete 100-point initialization records agree exactly. At the threshold, four changed words in failed return buffers are retained only as diagnostics. They are **not accepted coefficients** and receive no numerical accuracy, improvement or degradation comparison. Caller momentum/scale words remain unchanged and observed IEEE vectors agree, but neither is a witness of repaired or native reducer inputs. The mass correction does not resolve this stress failure.

The earlier pre-READY refusal and corrected pole-policy readiness check are preserved. Five later policy snapshots per arm retain the source-derived full-sum bypass and stored initialization flags; those flags are not learned filters. Native QP inputs, independent physical truth, genuine >53-bit physical masses, broader stress/cache coverage and promotion remain unqualified. No timings are added; Hua-Sheng's precise local performance acceptance remains pending.

Evidence: ordinary index `48a8b22cdb65…`, threshold index `df6f4e453f57…`. Exact identities and pure reproduction: `ew-qualification-selection.json`, `derive_evidence.py`. Style follows preserved pyAmpliCol revision `7266c1600f6e…`.

QCD $u\bar{u} \to Zggg$: five reducers

Descriptive timings on a shared host. This $2 \to 4$ study requests all five reducers; TIRust times out before READY. There is no CPU reservation or formal performance-acceptance claim. The preceding 44 pages are preserved.

GNU Fortran (GCC) 15.2.0; AMD EPYC 9754 128-Core Processor; affinity CPU 12; one-thread settings; shared, unreserved host. Exact flags/runtime hashes are retained.

CLI generation: `loop_sm; u u~ > z g g g [virt=QCD]`. Each traversal uses all **10 saved points** at $\sqrt{s} = 1000$ GeV. This pilot uses the unselected first ten of a 100-point fixture frozen before amplitudes; no failed point is removed. The main campaign remains separate. Ten frozen Latin/reversed cycles skip unavailable TIRust slots; four live positions are near-balanced. Two calibrations fix one full traversal per timed batch.

Table 1: Warmed full-amplitude wall time: median ± unscaled MAD. Ratios use matched cycles relative to NinjaRS8; bracketed 95% intervals are descriptive.

Reducer	READY s	Warmed ms/point median ± MAD	Ratio / 8	Accepted timed backend:precision calls	3e-6 / 1e-8
CutTools	41.65	3708 ± 118.6	×1.056 [1.023, 1.093]	1:DP=100	pass / pass
Ninja	36.91	3490 ± 81.71	×0.9715 [0.9505, 1.017]	6:DP=100	pass / miss
COLLIER	38.91	3950 ± 114.0	×1.111 [1.070, 1.127]	7:DP=100	pass / miss
NinjaRS	35.36	3564 ± 63.52	×1.000 [1.000, 1.000]	8:DP=100	pass / miss
TIRust	600 s timeout	unavailable	—	—	unqualified

Fortran SYSTEM_CLOCK wall time is divided by actual points × loops. The minimum whole-cohort batch exceeds the nominal 1.1 s target; no points are thinned. 95% intervals use 10,000 percentile bootstrap resamples of the median matched-cycle ratio (seed 20261112), resampling ten whole cycles; they do not test baseline acceptance. Actual timed batches span 33.64–41.91 s. READY includes the first amplitude and warmup, excluding startup and SETPARA. TIRust was cleanly terminated at its original 600 s READY budget; its initialization and warmed timings are unavailable.

Numerical results and precision

All four completed lanes pass MadLoop's 3e-6 regression criterion against same-source CutTools DP. Reducers 6/7/8 miss the separate 1e-8 target in finite terms at points 1, 2, 4, 5; the maximum raw relative residual is 4.2473e-08 at point 4. Overall/sole-order duplicate rows are not extra physical points. TIRust's incomplete ten-point lane is unqualified. Its point-4 QP finite value clusters with 6/7/8 (spread 4.765e-12); this neither supplies a timing nor establishes absolute truth or fault. Every miss and the original timeout remain retained.

All four timed lanes have native-DP accepted outputs. TIRust's partial warmup has 4 calls (9:DP=1, 9:QP=3); these are neither timed counts nor a complete ten-point traversal. Accepted counts do not expose internal retries or exception flags. Cards use full finite+UV+IR (LoopPoleMode=0), explicit COLLIER poles and normal stability/rescue. Four READY lanes have live DSO-map witnesses. TIRust9 has executable/loader/ldd/shared-file binding, with no startup live-map claim.

Hua-Sheng's local TODO remains: source-matched baseline/rescue-chain comparisons and upper-95% paired runtime-ratio bound ≤1.05 for every main process.

TIRust source: `10ce34ed8ad5b6f5f989a0e220eb3a244a71d1a1`; DSO SHA-256 (runtime scope described above): `85ee0cdf860e5ad4c7a5c26163802673de22404690a10c7c53fb73ebe1ff5bb0`. Original closed capsule: experiments/zggg-five-backend-timing-1; index SHA-256 e401cf5d4efb7708c277172eaa3be9f29bcadf54c5b3b484e7311109cf9a26cb. Reproduce with source binding, pure derivation and CSV. CSV retains full precision. Style: pyAmpliCol `7266c1600f6e…`; complete revision and attribution are in style-provenance.json.

Final source: five reducers and stock CLI

Initialization improves; warmed performance remains slower. At the diagnosed Zggg point TIRust/NinjaRS has matched-cycle median ratio ×**1.553**. The separate stock CLI gives ×1.605. Neither closes local runtime acceptance.

QCD $u\bar{u} \to Zggg$: one diagnosed physical point

Actual `loop_sm` CLI export; final source `77eb57b8`, library `d427`. This follow-up selects **point 1** of the original frozen 100-point set, after its initialization diagnosis. It is separate from the original ten-point timeout. Seven balanced cycles give 35 warmed full-amplitude batches, each one evaluation; actual batches span 3.190–5.712 s.

Table 1: Unscaled MAD of seven times; ratios are medians of matched within-cycle ratios, not ratios of medians. READY separately measures bootstrap plus one same-point warmup. No confidence interval is claimed.

Reducer	Median ± MAD ms / point	Paired ratio / NinjaRS 8	READY s	3e−6 / 1e−8
CutTools	3695.587 ± 57.175	×1.048	7.022	PASS / PASS
Ninja	3575.994 ± 84.260	×1.007	6.627	PASS / MISS
COLLIER	3837.506 ± 127.229	×1.099	7.497	PASS / MISS
NinjaRS	3526.731 ± 82.681	×1.000	6.775	PASS / MISS
TIRust	5426.217 ± 192.886	×1.553	14.119	PASS / MISS

All 45 retained comparisons pass `3e-6`. The separate `1e-8` target misses the finite coefficient for 6/7/8/9 versus CutTools by $\approx 3.534 \times 10^{-8}$ CutTools is its own reference. Old genuine-QP and the other reducers corroborate the cluster, without assigning automatic fault or absolute truth. No point is dropped.

Every reducer has seven accepted DP-transport dispatches. TIRust has **zero genuine Fortran-QP** dispatches but observed local tensor tiers. Timed primary routes are 7448 direct, 392 method-choice, 238 local106 and 42 local256; independent checks are 7518/350/182/70. Attempts106/256 are primary 252/42, check 182/70. Local512, refusal and unreported counts are zero. These counters exclude Fortran coefficient-cache hits. Local tiers lift 53-bit inputs with DP numerators; **219 is not a native-DP-only label**. Getter queries and writes are outside timing.

GNU Fortran 15.2.0; original model/math objects retained, usual generated sources without an optimization flag and polynomial source at O3 (exact recipes retained). Single CPU 12 affinity on a shared, unreserved Linux host; sampled one-minute load 288.1–314.9. This is a selected-point pilot, not a process average, full 100-point campaign or baseline-chain comparison.

Corrected stock `check profile`: QCD $u\bar{u} \to Z$

Table 2: One stock CLI run per backend; unequal short batches, unpaired and descriptive. FULL–SKIP estimates reduction cost; it is not a measured kernel time.

Reducer	FULL ms	SKIP ms	FULL−SKIP estimate ms	Points	FULL wall s
NinjaRS	0.077811	0.065435	0.012376	4117	0.341
TIRust	0.124896	0.064709	0.060187	1007	0.143

All configured helicities, full finite+UV+IR, pole mode 0 and CTModeRun 1. FULL now uses monotonic child-completion timing independent of RSS polling; SKIP/boot retain Python wall clocks. Stock values subtract boot time and divide by actual stage counts. Only the retained all-helicity end result establishes nonzero Born/finite; per-timed-point statuses and tensor routes are unavailable. Missing route counts are **not zero**. The configured polarized result is zero. Separate ten-point within-backend stability probes have 0 QP/unstable/exceptional; maximum spreads are 2.1200×10^{-14} / 1.6033×10^{-15}, not independent error. NinjaRS fresh initialization is 0.05830 s; TIRust reused its export, so initialization is **unavailable (null)**.

Initialization repair and retained precision limits

Final artifact and prototype timings have distinct identities. Final d427 passes 16 native tests, genuine 113⟶256 QP/poles/route smoke and 67 root integration tests. Route-aware tools add 50 tests and a 52-call tensor toy parity check, not another SM/performance qualification. Backend 9 stays explicit.

Quiet initialization: original versus adaptive prototype

The same original O0 Fortran executable, frozen point, cards, stability policy and CPU 12 affinity are used in one ordered old⟶prototype pair. No shim, preload or counter query; other owned captures were held. Shared CPUs were not reserved.

Table 3: Existing Fortran SYSTEM_CLOCK READY: bootstrap plus one same-point warmup, excluding launch and SETPARA. One descriptive pair, not warmed milliseconds/point or a confidence bound.

Library	READY s	Ratio / old	Outer result
Original signed S /85ee	255.241295	×1	323 · genuine Fortran QP
Adaptive prototype /470	15.345797	×0.06012	219 · DP inputs, local tensor tiers

The observed initialization ratio is **16.633**× old/prototype. Final d427 has its own 14.119 s READY above; the prototype timing is not relabeled. The original ten-point attempt's 600 s timeout remains on page 45.

Separate functional observer and tensor reference

The prototype observer overlapped background validation; its elapsed time is excluded. Of 4640 full-DP bridge requests (2320 primary / 2320 check), final routes are 4276 direct, 212 method-choice, 120 local106/DD, 32 local256; no 512/refusal/ unreported or genuine-QP entry. Attempts106/256 are primary 72/12, check 52/20; four 106 attempts continue to 256. Context-lifetime counters survive cache clears and omit pre-validation errors and Fortran coefficient-cache hits.

All comparisons with old true QP and reducers 1/6/7/8 pass `3e-6`; old QP and 6/7/8 also pass `1e-8`. Maximum candidate/old-QP residual is 1.6830×10^{-11}. The CutTools finite target miss 3.534×10^{-8} was already 3.532×10^{-8} against old QP. Genuine Fortran QP retains 113-bit input /256-bit work; local tiers use 53-bit inputs and a DP numerator. Numerical-body identity and final native qualification are recorded separately from prototype observations.

Eight exact-bit tensor fixtures compare against a **same-engine Float512 reference** on 53-bit inputs. All 176 degree×Laurent vectors pass `1e-8` (maximum 9.2988×10^{-17}); actual C-ABI fixture routes are 7 local106 + 1 local256. **50 individual component-relative misses remain** near UV zeros: absolute residue≤8.7664×10^{-27}, reference norm≤2.0021×10^{-146}. All 3136 complex differences are retained; no norm floor. Vector agreement is neither universal componentwise agreement nor an independent physics oracle.

Instrumented attribution; O2 remains unpromoted

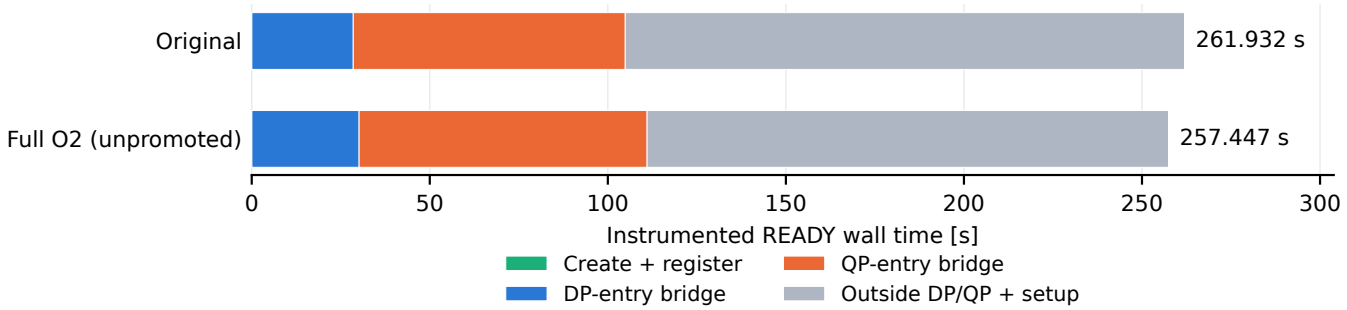


Figure 1: Original/O2 instrumented controls. Remainder is outside DP/QP+setup, not pure Fortran or clean kernel cost; it includes untraced/ boundary work, scheduling and observer overhead.

Original READY 261.932 s contains setup 0.039, DP-entry 28.468, QP-entry 76.383, remainder 157.041 s; first DP⟶QP gap 69.761 s. Its 152 DP refusals lead to whole-QP retries; the separate functional observer accepts 152 local tensor rescues. This attribution is not a phase decomposition of the quiet pair. Full-O2 READY is 257.447 s (×0.983); literal outputs and metadata/status fingerprints match, without full-input proof. No convincing single-control benefit; O2 is unpromoted.

Acceptance remains open. All 66 fresh source-matched numerical lanes and eight main runtime gates await Hua-Sheng's local run. TIRust remains slower in both measured warmed comparisons above. The upper 95% paired candidate/ baseline bound≤1.05 for every main process is not established. Stress failures, strict-target misses, old coarse CLI controls and original timeout remain retained.

Current source 77eb57b82d48 / final DSO d427667a0f2f; prototype 607e7f58842b /470d46df9c8f. Instrumented MG cd014592; old stock cafb35d7 /85ee remains separately archived. Signed TIRust 10ce34ed8ad5; exact hashes, nine CSVs, raw points/checks/routes and pure reproduction accompany this appendix. Original 16 MB source archive is reused by hash. Typography/palette: pyAmpliCol 7266c1600f6e, preserved style.

Stock O3 profile: QCD $u\bar{u} \to Zggg$

A separate stock-profile measurement. Actual `loop_sm` exports use O3 for all generated MODEL/DHELAS/process/driver objects, normal learned filters and DP-interface timing without the independent stability test. Earlier checked, fixed-point O0 results keep their own input/filter policy; ratios across these pages are not compiler speedups.

Five explicit reduction backends

Table 1: Actual stock `check profile` per-point estimates. Each row is a separate run with its own batch size; FULL–SKIP estimates reduction cost, not a directly measured kernel time. Unavailable/failed results remain explicit.

Reducer	FULL ms	SKIP ms	FULL−SKIP estimate ms	FULL points	FULL wall s	FULL ratio / NinjaRS 8
CutTools	1579.818810	1458.729219	121.089591	50	79.133	×1.089
Ninja	1504.547799	1451.542921	53.004877	50	75.357	×1.037
COLLIER	1771.917629	1406.543589	365.374041	50	88.729	×1.221
NinjaRS	1450.696761	1437.618103	13.078658	50	72.661	×1.000
TIRust	2260.844975	1368.611798	892.233177	50	113.179	×1.558

TIRust FULL is 2260.845 ms versus NinjaRS 1450.697 ms: ratio 1.55845, or 55.8% higher. This remaining runtime gap is explicit; the ratio is descriptive, with no paired confidence interval.

Measured configuration and clock

`CTModeRun=1`, `LoopPoleMode=-1`, `HelicityFilterLevel=2`; normal filter learning is retained. COLLIER's UV and IR controls are both true. These effective cards, selected helicities/filter state, model and all-generated-object O3 compilation records are retained with the raw logs. Installed reducer release builds are unchanged. Timed CTMode1 omits MadLoop direction/rotation checks and whole-amplitude QP rescue. All runs retain 48 contributing helicities out of 96. Timed energy is 1 TeV, verified from the actual FULL driver, external masses and physical card. GNU Fortran 15.2.0; CPU 12 affinity on a shared, unreserved host. Observed O3 compiler commands cover 174 base objects; quiet stock recompiles inherit O3 make_opts/environment/recipe, without directly logged argv. The initial model dependency-race build and successful retry are both retained.

FULL is the log's helicity-sum time per phase-space point: (monotonic child-completion wall minus stock boot time)/actual count. SKIP and boot retain Python wall clocks. Each FULL and SKIP stage contains 50 points. Stock/filter initialization uses a fresh three-point learning startup; compiler time is separate. It is not a library-only microbenchmark or the earlier diagnosed one-point initialization experiment.

Table 2: Initialization is separate from warmed stage estimates. A null initialization denotes unavailable measurement. Accepted DP transport does not establish native-DP arithmetic for TIRust.

Reducer	Initialization s	Status	Observed precision scope
CutTools	47.64849	Completed; unchecked 110	Configured DP; timed routes unknown
Ninja	19.36933	Completed; unchecked 110	Configured DP; timed routes unknown
COLLIER	113.52161	Completed; unchecked 110	Configured DP; timed routes unknown
NinjaRS	16.89482	Completed; unchecked 110	Configured DP; timed routes unknown
TIRust	94.28168	Completed; unchecked 110	Configured DP; timed routes unknown

Accuracy, scope and remaining acceptance

One retained common FULL endpoint has identical six-momentum and normalization words across all five runs. Its ten Born/finite/pole pairwise comparisons pass 3e-6 and 1e-8 (maximum relative residual 1.759e-09). This does not qualify all 50 timed points. Endpoint return code 110 is unchecked and output freshness was not independently instrumented. The separate one-point stability phase is outside timing.

TIRust's DP-input path may use local 106-bit DD or 256/512-bit tensor arithmetic with DP inputs/numerators. Those tiers differ from genuine 113-bit Fortran-QP input transport. Stock timings alone do not assign per-point tensor routes; missing counts must not be read as zero. These are configured-backend timings, not accepted-backend or native-DP-only status proofs.

These are unpaired shared-host stock runs, with no paired confidence interval or main-process acceptance claim. All 66 fresh local numerical lanes and eight main runtime gates remain pending. The upper 95% paired candidate/baseline runtime bound ≤1.05 for every main process is not established. The integration goal remains incomplete.

Source `cda704e28d634e893fabc3ac358d9a27789f93aa` TIRust DSO `d427667a0f2f55d0bd96778c1259ffcf52e3925341890b81c38f5f94e1eb7052`. Exact FULL/SKIP/boot counts, clocks, filters, compiler recipes, failures and initialization records accompany `stock-o3-timings.csv` and the pure reproduction commands. The preceding 47 pages are preserved transitively, without replaying or relabeling their physics. Style: pyAmpliCol 7266c1600f6e, preserved typography and green–white–red timing palette.

Current-source requalification: TIRust pin `9a781998`

Rehearsal on an unmerged pin. These pages qualify MadGraph `624ef69f05` with TIRust `9a781998e54a39fb06d3621de3cd458da09d2a52`, the head of TIRust PR 35 (not yet on TIRust `main`). The final pin must be a merged `main` commit; the same commands are rerun on it. They supersede the earlier pages only for this source and runtime (DSO `6c11cfd9eecf`). Shared-host results; precise paired timing acceptance remains Hua-Sheng's local TODO.

The fresh `install tirust` build of this pin passes `smoke --require-poles`, the pinned upstream C-ABI fixtures (27 DP and 54 QP lanes) and links only `libgcc_s`, `libm` and `libc`. Exports were regenerated from the campaign's frozen cards, models and `-O3` flags; the baseline is the starting revision 9228fc58.

Ordinary matrix: 11 processes × 6 lanes × 100 points

Table 1: Born, finite and both poles against starting-revision CutTools (`LoopMEComparator`). 66 of 66 lanes pass `3e-6`; 54 meet the separate `1e-8` target. Misses count scalar comparisons (split-order rows duplicate). "Previous" is source 0fcd2cd873 / runtime 854b79dc; the earlier EW 2⟶4 TIRust lane (40 DP / 60 QP) was recorded separately.

Process	3e-6	1e-8 misses CT/Ninja/COLLIER/ NinjaRS/TIRust	TIRust max finite residual	TIRust DP/QP	previous DP/QP
QCD u ū → Z	6/6	0/0/0/0/0	5.4e-14	100/0	100/0
QCD g g → t t̄	6/6	0/0/0/0/0	2.8e-12	100/0	100/0
QCD u ū → Z g g	6/6	0/56/56/56/56	2.8e-6	100/0	87/13
QCD u ū → Z Z g g	6/6	0/88/80/80/78	4.1e-7	100/0	14/86
EW u d̄ → W⁺	6/6	0/0/0/0/0	9.4e-13	100/0	100/0
EW e⁺e⁻ → μ⁺μ⁻	6/6	0/0/0/0/0	2.3e-13	100/0	100/0
EW e⁺e⁻ → t t̄ γ	6/6	0/0/0/0/0	1.1e-9	100/0	100/0
EW e⁺e⁻ → μ⁺μ⁻ γ γ	6/6	0/12/10/14/8	1.1e-7	99/1	–
QCD+QED u ū → g γ	6/6	0/0/0/0/0	2.6e-11	100/0	100/0
loop-induced g g → H	6/6	0/0/0/0/0	1.5e-10	100/0	100/0
loop-induced g g → H H	6/6	0/0/0/0/0	6.4e-10	100/0	100/0

TIRust now accepts double precision where the previous runtime needed a genuine QP rescue: its internal double-double and MPFR tensor tiers recover refused lanes before MadLoop rebuilds a QP numerator. In QCD $u\bar{u} \to Zgg$, Ninja, COLLIER, NinjaRS and TIRust miss the `1e-8` target on the same 28 points with the same residual, which points at the CutTools reference; the comparison itself assigns no independent truth.

Double-double gate (headroom 26)

Table 2: Every bridge call of the TIRust lane was captured (`tools/tirust_capture`) and replayed: the production build reproduces all recorded bits; references with headroom 40 (the previous 2^−66 gate) and 105 (no lane in DD) evaluate the same lanes at 256 bits. Error = max |difference| / max |value| of the same tensor degree. Flag 1e-12, fail 1e-9. 6131 DD calls in total.

Process	DP calls	DD calls	new-window calls	max error new window	max error all DD	verdict
EW u d̄ → W⁺	1212	0	0	—	—	not exercised
EW e⁺e⁻ → μ⁺μ⁻	5252	0	0	—	—	not exercised
EW e⁺e⁻ → t t̄ γ	40400	0	0	—	—	not exercised
EW e⁺e⁻ → μ⁺μ⁻ γ γ	246036	1545	60	1.9e-24	1.9e-24	qualified
loop-induced g g → H	808	0	0	—	—	not exercised
loop-induced g g → H H	3232	0	0	—	—	not exercised
QCD+QED u ū → g γ	5656	0	0	—	—	not exercised
QCD u ū → Z	202	0	0	—	—	not exercised
QCD g g → t t̄	5656	0	0	—	—	not exercised
QCD u ū → Z g g	23432	134	8	1.9e-27	3.6e-24	qualified
QCD u ū → Z Z g g	150490	4452	414	2.0e-22	1.9e-20	qualified

Per-point arithmetic provenance

Table 3: Each of the 100 ordinary points per process evaluated once by a private clone with bridge route counters around every amplitude call (`tools/tirust_provenance.py`). MadLoop clears the TIRust and Fortran coefficient caches per point, so the deltas cover every reduction a point reused. Cache-off and a 1 MiB budget (insertion refusals) reproduce every coefficient, accuracy and status bit for bit.

Process	binary64 only	DD tier	MPFR tier	genuine QP	failed	status = campaign	cache off identical	1 MiB budget identical
EW u d̄ → W⁺	100	0	0	0	0	100/100	yes	yes
EW e⁺e⁻ → μ⁺μ⁻	100	0	0	0	0	100/100	yes	yes
EW e⁺e⁻ → t t̄ γ	100	0	0	0	0	100/100	yes	yes
EW e⁺e⁻ → μ⁺μ⁻ γ γ	40	58	1	1	0	100/100	yes	yes
loop-induced g g → H	100	0	0	0	0	100/100	yes	yes
loop-induced g g → H H	100	0	0	0	0	100/100	yes	yes
QCD+QED u ū → g γ	100	0	0	0	0	100/100	yes	yes
QCD u ū → Z	100	0	0	0	0	100/100	yes	yes
QCD g g → t t̄	100	0	0	0	0	100/100	yes	yes
QCD u ū → Z g g	87	13	0	0	0	100/100	yes	yes
QCD u ū → Z Z g g	14	71	15	0	0	100/100	yes	yes

Physical stress points

Table 4: Threshold, soft/collinear, small-Gram and complex-mass points (`tools/tirust_stress.py`). Reference: forced-QP CutTools and Ninja when they agree, otherwise stability-checked values of independent reducers that agree. TIRust may refuse; only an accepted value that disagrees would block (0 such cases). Refusals where every reducer fails are properties of the point. “Process STOP”: CutTools ends the program (CT692) on the ZZgg soft-collinear point; on the most forward CMS small-Gram point it never returns. Any lane falling back to CutTools therefore aborts or hangs there; the driver records the point and resumes with the next one. “TIRust QP” compares unchecked forced QP (`CTModeRun=4`) with the reference. “differs” occurs only where the genuine-QP interface refuses tensors (QP failure latch set; e.g. 48 rank-4 boxes at the H H small-Gram point that the DP-input path with its 512-bit tier evaluates correctly); automatic mode never accepts such results.

Process	point	reference	TIRust alone	chain 9|6|7|1	TIRust QP
CMS EW e⁺e⁻ → μ⁺μ⁻	12 ordinary	QP CutTools = QP Ninja ×12	accepted-agrees (219) ×12	accepted-agrees (219) ×12	agrees ×12
CMS EW e⁺e⁻ → μ⁺μ⁻	z-resonance	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
CMS EW e⁺e⁻ → μ⁺μ⁻	z-resonance	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
CMS EW e⁺e⁻ → μ⁺μ⁻	z-resonance	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
CMS EW e⁺e⁻ → μ⁺μ⁻	internal-ww-threshold	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
CMS EW e⁺e⁻ → μ⁺μ⁻	internal-ww-threshold	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
CMS EW e⁺e⁻ → μ⁺μ⁻	internal-ww-threshold	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
CMS EW e⁺e⁻ → μ⁺μ⁻	small-gram	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
CMS EW e⁺e⁻ → μ⁺μ⁻	small-gram	checked CutTools	refused (420)	rescued-by-1-agrees (329)	differs
CMS EW e⁺e⁻ → μ⁺μ⁻	small-gram	unavailable or disputed	refused (420)	unrescued (process hang) (–)	–
EW e⁺e⁻ → μ⁺μ⁻	small-gram	checked CutTools	refused (420)	rescued-by-1-agrees (329)	differs
EW e⁺e⁻ → t t̄ γ	threshold	checked results disagree	refused (420)	rescued-by-6-unreferenced (328)	–
EW e⁺e⁻ → t t̄ γ	soft-collinear	unavailable or disputed	refused (420)	unrescued (420)	–
EW e⁺e⁻ → t t̄ γ	small-gram	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
EW e⁺e⁻ → μ⁺μ⁻ γ γ	soft-collinear	unavailable or disputed	refused (420)	unrescued (420)	–
EW e⁺e⁻ → μ⁺μ⁻ γ γ	small-gram	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
loop-induced g g → H H	threshold	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees

Process	point	reference	TIRust alone	chain 9|6|7|1	TIRust QP
loop-induced g g → H H	small-gram	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	differs
QCD+QED u ū → g γ	small-gram	unavailable or disputed	refused (420)	unrescued (420)	–
QCD g g → t t̄	threshold	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
QCD g g → t t̄	small-gram	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
QCD u ū → Z g g	threshold	unavailable or disputed	refused (420)	unrescued (420)	–
QCD u ū → Z g g	soft-collinear	unavailable or disputed	refused (420)	unrescued (420)	–
QCD u ū → Z g g	small-gram	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees
QCD u ū → Z Z g g	threshold	unavailable or disputed	refused (420)	unrescued (420)	–
QCD u ū → Z Z g g	soft-collinear	unavailable or disputed	refused (420)	unrescued (process STOP) (–)	–
QCD u ū → Z Z g g	small-gram	QP CutTools = QP Ninja	accepted-agrees (219)	accepted-agrees (219)	agrees

Stability (Appendix C of arXiv:1604.01363)

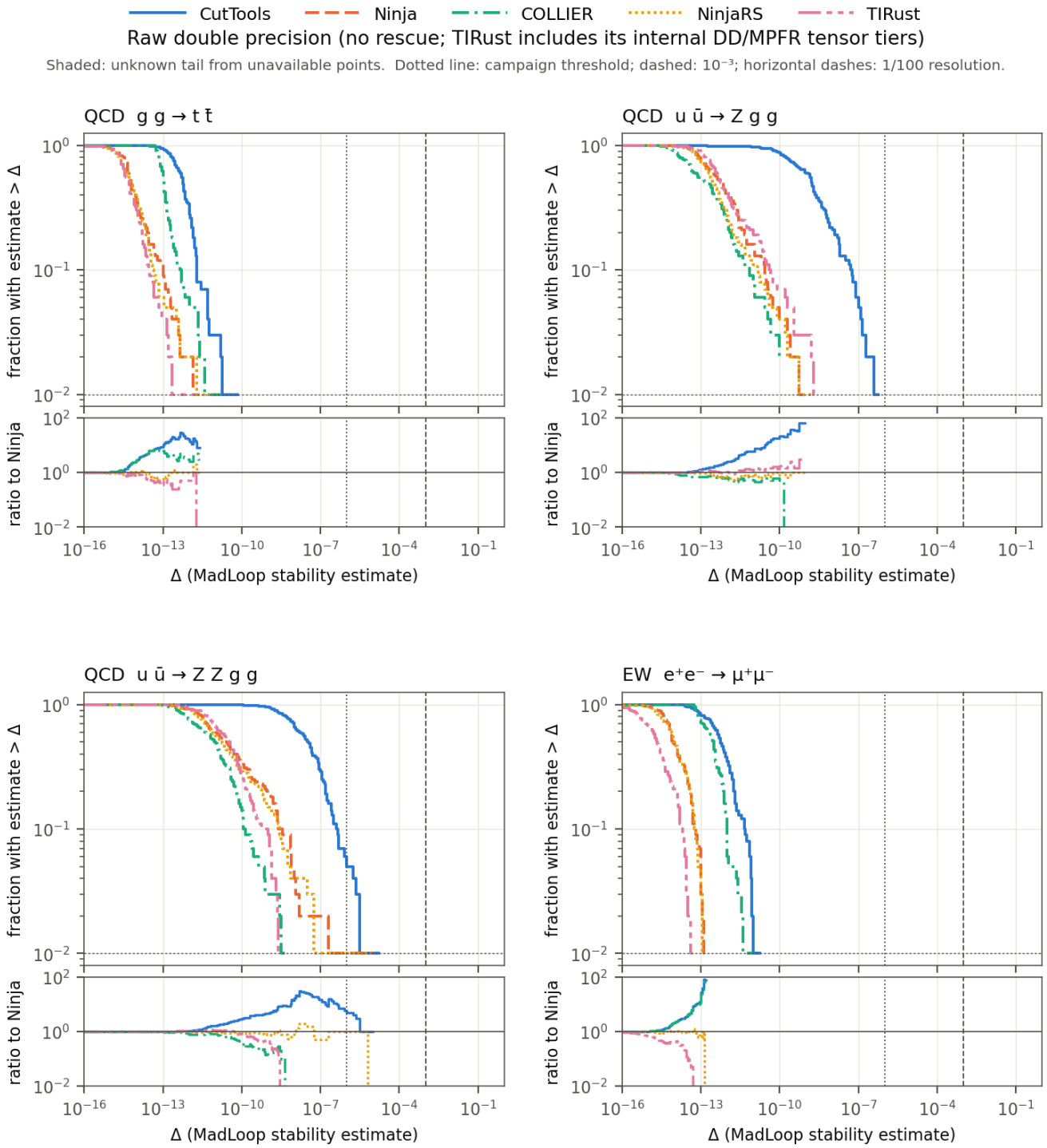


Figure 1: Raw double precision, no automatic rescue: MadLoop's internal estimate of the total answer on 100 frozen cut-qualified points per process. Shaded bands are unknown tails from unavailable points; a TIRust point is unavailable when its reduction refuses. TIRust's DP includes its internal DD/MPFR tensor tiers.

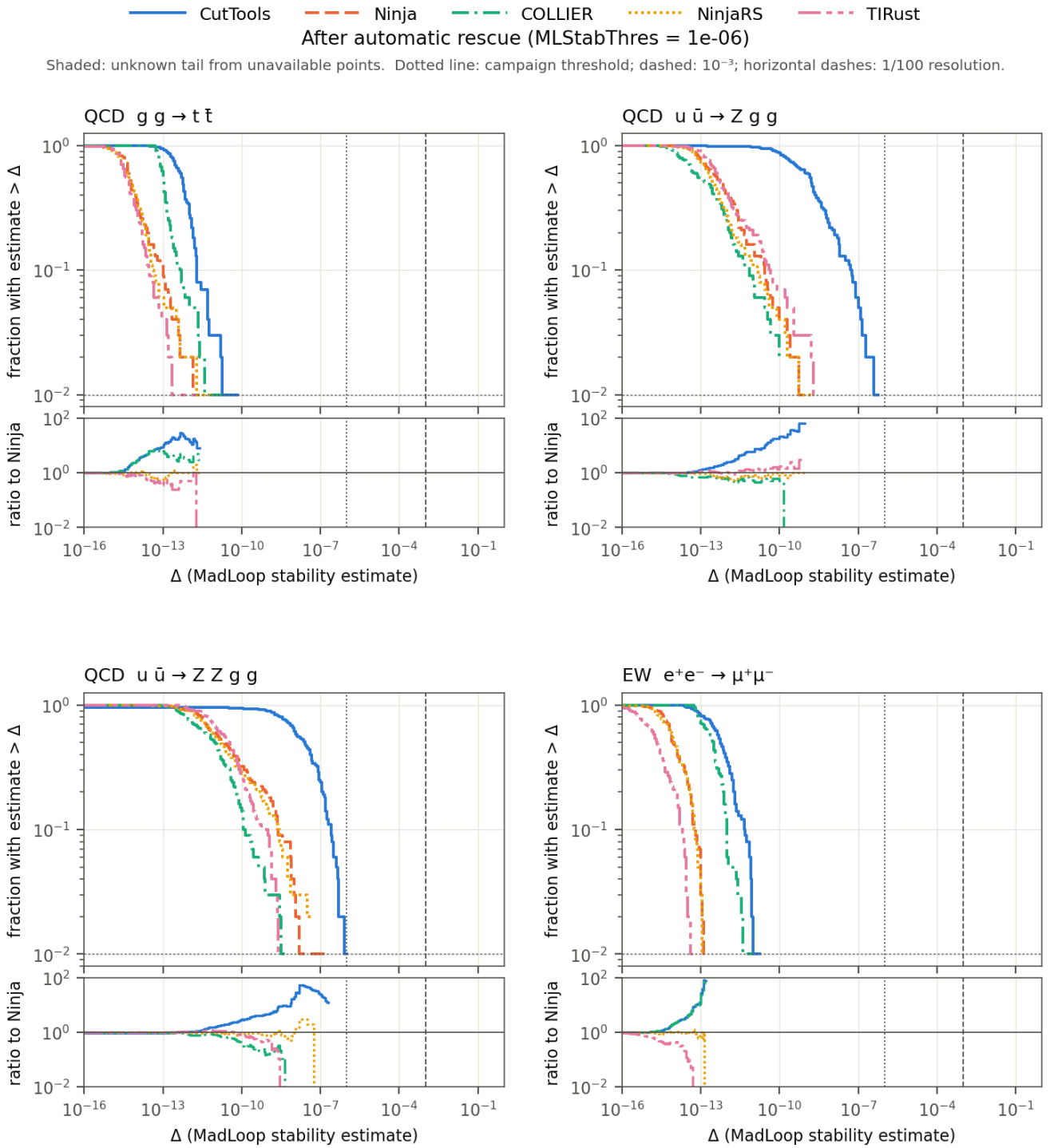


Figure 2: After MadLoop's automatic rescue at the campaign threshold `MLStabThres = 1e-6`.

Table 5: Fractions of the 100 points whose estimate exceeds the threshold; brackets give [known, known + unavailable] bounds. At 100 points the resolution is 0.01; larger samples remain a TODO. The estimate is MadLoop's self-consistency test, not an independent error.

Variant	Process	Reducer	available	> 1e-3	> 1e-6	> 1e-9
raw	QCD g g → t t̄	CutTools	100/100	0.00	0.00	0.00
raw	QCD g g → t t̄	Ninja	100/100	0.00	0.00	0.00
raw	QCD g g → t t̄	COLLIER	100/100	0.00	0.00	0.00
raw	QCD g g → t t̄	NinjaRS	100/100	0.00	0.00	0.00
raw	QCD g g → t t̄	TIRust	100/100	0.00	0.00	0.00
raw	QCD u ū → Z g g	CutTools	100/100	0.00	0.00	0.60
raw	QCD u ū → Z g g	Ninja	100/100	0.00	0.00	0.00
raw	QCD u ū → Z g g	COLLIER	100/100	0.00	0.00	0.00
raw	QCD u ū → Z g g	NinjaRS	100/100	0.00	0.00	0.01

Variant	Process	Reducer	available	> 1e-3	> 1e-6	> 1e-9
raw	QCD u ū → Z g g	TIRust	100/100	0.00	0.00	0.03
raw	QCD u ū → Z Z g g	CutTools	100/100	0.00	0.05	0.95
raw	QCD u ū → Z Z g g	Ninja	100/100	0.00	0.01	0.20
raw	QCD u ū → Z Z g g	COLLIER	100/100	0.00	0.00	0.03
raw	QCD u ū → Z Z g g	NinjaRS	100/100	0.00	0.01	0.15
raw	QCD u ū → Z Z g g	TIRust	100/100	0.00	0.00	0.09
raw	EW $e^+e^- \to \mu^+\mu^-$	CutTools	100/100	0.00	0.00	0.00
raw	EW $e^+e^- \to \mu^+\mu^-$	Ninja	100/100	0.00	0.00	0.00
raw	EW $e^+e^- \to \mu^+\mu^-$	COLLIER	100/100	0.00	0.00	0.00
raw	EW $e^+e^- \to \mu^+\mu^-$	NinjaRS	100/100	0.00	0.00	0.00
raw	EW $e^+e^- \to \mu^+\mu^-$	TIRust	100/100	0.00	0.00	0.00
rescued	QCD g g → t $\bar{t}$	CutTools	100/100	0.00	0.00	0.00
rescued	QCD g g → t $\bar{t}$	Ninja	100/100	0.00	0.00	0.00
rescued	QCD g g → t $\bar{t}$	COLLIER	100/100	0.00	0.00	0.00
rescued	QCD g g → t $\bar{t}$	NinjaRS	100/100	0.00	0.00	0.00
rescued	QCD g g → t $\bar{t}$	TIRust	100/100	0.00	0.00	0.00
rescued	QCD u ū → Z g g	CutTools	100/100	0.00	0.00	0.60
rescued	QCD u ū → Z g g	Ninja	100/100	0.00	0.00	0.00
rescued	QCD u ū → Z g g	COLLIER	100/100	0.00	0.00	0.00
rescued	QCD u ū → Z g g	NinjaRS	100/100	0.00	0.00	0.01
rescued	QCD u ū → Z g g	TIRust	100/100	0.00	0.00	0.03
rescued	QCD u ū → Z Z g g	CutTools	100/100	0.00	0.00	0.90
rescued	QCD u ū → Z Z g g	Ninja	100/100	0.00	0.00	0.19
rescued	QCD u ū → Z Z g g	COLLIER	100/100	0.00	0.00	0.03
rescued	QCD u ū → Z Z g g	NinjaRS	100/100	0.00	0.00	0.14
rescued	QCD u ū → Z Z g g	TIRust	100/100	0.00	0.00	0.09
rescued	EW $e^+e^- \to \mu^+\mu^-$	CutTools	100/100	0.00	0.00	0.00
rescued	EW $e^+e^- \to \mu^+\mu^-$	Ninja	100/100	0.00	0.00	0.00
rescued	EW $e^+e^- \to \mu^+\mu^-$	COLLIER	100/100	0.00	0.00	0.00
rescued	EW $e^+e^- \to \mu^+\mu^-$	NinjaRS	100/100	0.00	0.00	0.00
rescued	EW $e^+e^- \to \mu^+\mu^-$	TIRust	100/100	0.00	0.00	0.00

Provisional stock O3 profile (not acceptance)

Table 6: Stock `check profile` of QCD $u\bar{u} \to Zggg$ on the existing O3 export (learned filters, 48/96 helicities, both poles, timed `CTModeRun=1` without stability test), 50 points, one CPU, three repetitions in alternating reducer order, TIRust DSO `6c11cfd9eecf`. TIRust/NinjaRS 1.177 (the 48-page report measured 1.558 with the previous runtime). Unpaired shared-host data: no confidence bound, no acceptance claim.

Reducer	FULL ms/point (median)	three repetitions	initialization s	ratio / NinjaRS
CutTools	1028.5	1021.5 / 1040.9 / 1028.5	32.2	1.069
Ninja	968.2	970.2 / 968.2 / 967.2	12.8	1.007
COLLIER	1183.5	1183.5 / 1189.4 / 1176.3	73.5	1.230
NinjaRS	961.9	978.2 / 961.9 / 958.6	11.8	1.000
TIRust	1128.1	1151.1 / 1126.8 / 1128.1	8.0	1.173

Campaign `tirust-9a781998-1` (manifest `bb8b530e57a6`), exporter `be13df837e4f`. Commands, summaries and the local timing handoff are in `docs/benchmarkfolder/campaigns/tirust-9a781998-1/`. The preceding 48 pages are unchanged.